

Construction ERP

Complete User Manual

Web Portal · Admin Console · Mobile App · Marketing site

Every screen and tab — with the complete data flow from UI to API to database.

Planning → execution → completion: progress, workforce, materials, BOQ, quality, HR & payroll, sales, equipment, audit.

Document	User Manual — complete screen guide with full data flow
Version	2.0 (supersedes 1.0)
Product	BuildTrack — Construction Project Tracking · v1.0
Covers	Web Portal (web-app), Admin Console (admin-panel), Mobile App (mobile), Marketing site, Backend API & MySQL data model
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This document was generated from the source code of the Construction ERP repository (nitesh1414/Contruction-ERP). Wherever a screen description and the code can diverge, the code in ``backend/``, ``web-app/``, ``admin-panel/``, ``mobile/`` and ``marketing/`` is the source of truth.

Contents

1. System overview	4
1.1 Key concepts you must know first	4
1.2 Architecture & data flow (big picture)	4
2. Roles, permissions & access control	7
2.1 Permission model	7
2.2 Seeded roles	7
2.3 Demo logins (after `npm run db:setup`)	7
2.4 How access control appears in the UI	8
3. Logging in & account basics	9
3.1 Sign in (Web Portal and Admin Console)	9
3.2 Forgot password	9
3.3 My Profile (Web Portal)	9
4. Web Portal — complete screen & tab guide	11
4.0 Global chrome: top bar, search, notifications bell, user menu	11
4.1 Dashboard	11
4.2 Projects	12
4.3 Project Details	13
4.4 Daily Progress	14
4.5 Workforce & Attendance	16
4.6 Equipment	18
4.7 Tasks & Milestones	19
4.8 Drawings	20
4.9 Inspections	21
4.10 Test Reports (quality)	22
4.11 Documents	23
4.12 Issues / Snags	24
4.13 Materials & Inventory	25
4.14 BOQ (Bill of Quantities)	30
4.15 Payments (Billing & cost entries)	32
4.16 Petty Cash	33
4.17 CRM & Sales	34
4.18 HR & Payroll	35
4.19 Reports & Dashboards	41
4.20 My Profile	43
4.21 In-app notifications (Web Portal)	43
5. Admin Console — complete screen & tab guide	44
5.0 Layout & navigation	44
5.1 Admin Dashboard	44
5.2 Users & Employees	45
5.3 Roles & Permissions	46
5.4 Projects & Wings (admin)	48
5.5 Masters (11 tabs)	49
5.6 HR & Payroll (admin operations view)	50

5.7 Petty Cash (admin view)	51
5.8 Audit Logs	51
5.9 Broadcast	51
6. Mobile App — screen guide (site engineer)	52
6.1 Screens	52
6.2 Offline queue (how it works)	52
6.2b Marketing site	52
7. End-to-end data flows (complete)	53
7.1 Authentication & session lifecycle	53
7.2 User & employee provisioning (identity sync)	53
7.3 Project setup & progress roll-up	53
7.4 Daily progress entry — web & mobile (with photos, GPS, offline)	53
7.5 Material procurement cycle (requirement → PO → GRN → stock)	54
7.6 Material consumption on site	54
7.7 Workforce wage cycle (worker → attendance → payment)	54
7.8 HR payroll cycle (structure → leave → bulk payroll → payment)	55
7.9 Unit sales & collections cycle	55
7.10 Equipment operation & hire billing cycle	55
7.11 Quality assurance flow	55
7.12 Issue lifecycle & notifications	56
7.13 Drawing revision control	56
7.14 Audit trail & broadcasts (system-wide)	56
8. Data dictionary — all 65 database tables	57
8.1 Identity & access (9)	57
8.2 Project structure & progress (10)	57
8.3 Materials & inventory (11)	57
8.4 Finance (6)	57
8.5 Workforce (5)	58
8.6 BOQ (3)	58
8.7 Quality & issues (8)	58
8.8 Documents, equipment & system (6)	58
8.9 HRMS (7)	58
9. Status & workflow reference	60
10. Tips, FAQ & troubleshooting	61

How to read this manual: Every screen section follows the same pattern: what the screen is for, how to operate it step by step, and a **Data flow** table tracing each user action from the UI, through the REST API endpoint, to the exact MySQL tables written or read, including side effects (notifications, stock updates, status changes, audit entries). Chapter 7 combines these into the complete cross-module data flows a site team actually runs day to day.

1. System overview

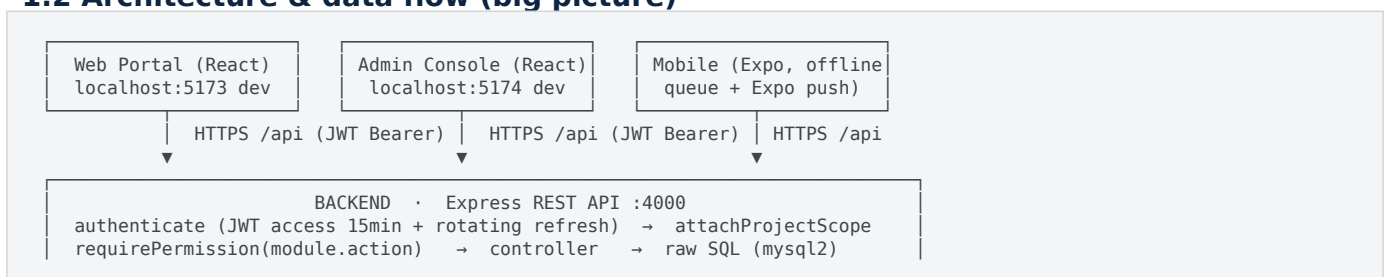
Construction ERP is a complete project-tracking platform for construction companies. It is built from five deployable applications that all talk to one backend and one MySQL database:

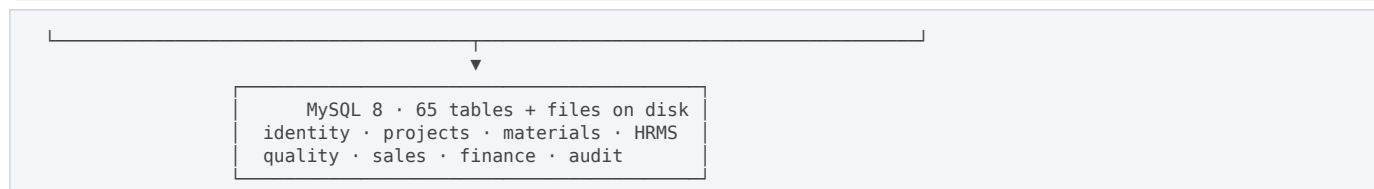
App	Stack	Who uses it	Purpose
Backend API (backend/)	Node.js 18+, Express, raw mysql2 (no ORM)	—	REST API: JWT auth, RBAC, project/wing scoping, file uploads, audit trail, 65-table schema
Web Portal (web-app/)	React 18 + Vite + TypeScript	PMs, engineers, store, sales, accounts, quality & safety	Day-to-day project work: progress, workforce, materials, BOQ, quality, HR & payroll, petty cash, sales, reports
Admin Console (admin-panel/)	React 18 + Vite + TypeScript	Super Admin / Admin	Users & employees, roles & permission matrix, master data, audit logs, broadcasts
Mobile App (mobile/)	React Native + Expo SDK 54	Site engineers & field staff	Offline-first daily progress with camera + GPS, attendance, issues, inspections, sync queue, push notifications
Marketing site (marketing/)	React 18 + Vite + TypeScript	Visitors	Public landing page describing the ERP's actual capabilities

1.1 Key concepts you must know first

- **Project structure.** Every project is organised as **Projects → Wings/Blocks → Floors → Units**. This hierarchy drives progress roll-up, access scoping and reporting.
- **Automatic progress roll-up.** When a daily progress report is saved, the server recomputes: **wing progress** = average of all `daily_progress.percentage` rows for that wing; **project overall progress** = average of its wings' progress. You never type project % by hand.
- **Role-based access (RBAC).** Every module has up to 8 permission actions: `view · create · edit · delete · approve · export · upload · download`. Roles bundle permissions; users get roles. The UI hides menu items and buttons you cannot use — and the API re-checks every single request.
- **Project & wing scoping.** Beyond module permissions, each user is assigned specific **projects (and optionally wings)** in `user_projects`. Scoped users only see and act on their own projects. A row with `wing_id = 0` means "all wings of that project". Super Admin and the Admin role are global.
- **One shared employee directory.** The Admin Console and the Web Portal read and write the **same** `hrms_employees` records. A login account and its employee record represent the same person, linked by `hrms_employees.user_id`. You can create either side and offer to create/link the other; the link is one-to-one and never bypasses RBAC.
- **Everything is audited.** Every create / update / delete / export writes a row to `audit_logs` with who, when, which module and record, the IP, and the full old/new JSON values.
- **Event-driven notifications.** Assignments, approvals, rejections, low stock, overdue issues, failing tests, expiring documents and pending collections raise in-app notifications (and Expo push on mobile). Every event can be switched off in Admin → Masters → Notification Events.
- **Offline-first mobile.** The mobile app queues progress reports and issues locally (with a unique `client_ref`) when there is no network and auto-syncs in the background. The server de-duplicates by `client_ref`, so re-syncing never creates double entries.

1.2 Architecture & data flow (big picture)





Data-flow rule of thumb for every screen in this manual:

UI action → one REST call → controller validates (permission + project scope) → transactional SQL writes the business tables → side effects (stock ledger, progress roll-up, notifications, audit row) → JSON response → UI refreshes.

2. Roles, permissions & access control

2.1 Permission model

- A **permission** is a code `module.action`, e.g. `materials.approve`, `hrms.create`, `attendance.export`.
- A **role** is a named set of permission ids (`role_permissions` join table).
- A **user** has one or more **roles** (`user_roles`) and an optional list of **project/wing assignments** (`user_projects`).
- The **Super Admin** role (`super_admin`) bypasses every check. The **Admin** role (`admin`) is global across projects.
- All other users must hold the specific `module.action` permission **and** be inside their project/wing scope for any record-level operation.

Permission codes are managed centrally (Admin Console → Roles → Permissions matrix); custom roles can be created at any time from any combination of checkboxes.

2.2 Seeded roles

The installer seeds 12 ready-made roles. Admins can create further **custom** roles.

Role (code)	Typical daily work
Super Admin (<code>super_admin</code>)	Full unrestricted access to both apps, all data, role matrix
Admin (<code>admin</code>)	System administration: users, masters, audit, broadcasts (global project visibility)
Project Manager (<code>project_manager</code>)	Owns assigned projects: structure, milestones, approvals (requirements, POs, BOQ), reports, billing, workforce
Site Engineer (<code>site_engineer</code>)	Daily progress with GPS photos, attendance, issues, inspections, consumption — on assigned projects/wings
Civil / Electrical / Plumbing Engineer (<code>civil_engineer</code> , <code>electrical_engineer</code> , <code>plumbing_engineer</code>)	Same as Site Engineer, scoped to their discipline (demo accounts are wing-scoped to demonstrate scoping)
Sales Executive (<code>sales_executive</code>)	Unit sales, booking-to-payment tracking, collections, sales export
Store Manager (<code>store_manager</code>)	Material master, requirement → PO → GRN → stock, suppliers, returns, low-stock follow-up
Accountant (<code>accountant</code>)	Cost/billing entries, vendor dues, labour payments, payroll payments, financial reports
Safety Officer (<code>safety_officer</code>)	Safety inspections, safety issues, documents
Quality Engineer (<code>quality_engineer</code>)	Material test reports (approve/reject), inspection checklists
Custom roles	Anything you build — e.g. "Contractor (view-only)", "Client (view-only)", "Surveyor"

2.3 Demo logins (after `npm run db:setup`)

All demo accounts exist in the seeded database. **Change all passwords before any real deployment.**

Role	Email	Password
Super Admin	admin@constructionerp.com	Admin@123
Admin (Console)	admin.user@constructionerp.com	Password@123
Project Manager	pm@constructionerp.com	Password@123
Site Engineer	engineer@constructionerp.com	Password@123
Civil / Electrical / Plumbing Engineer	civil@ / electrical@ / plumbing@constructionerp.com	Password@123
Sales Executive	sales@constructionerp.com	Password@123
Store Manager	store@constructionerp.com	Password@123
Accountant	accounts@constructionerp.com	Password@123
Safety Officer	safety@constructionerp.com	Password@123
Quality Engineer	quality@constructionerp.com	Password@123

2.4 How access control appears in the UI

- **Web Portal sidebar** items and **Admin Console** menu items are filtered by `canAny(perm)` — if your roles have none of the required permission for a module, the menu entry simply does not exist.
- **Route guard.** Opening a URL directly without the required permission shows a "No access — contact your administrator" card.
- **Row-level buttons.** Approve / Reject / Delete / Export buttons appear only when the matching `module.action` permission is present.
- **Server-side enforcement.** Even with a tampered client, the API returns **403 Missing permission:** `module.action` or **403 No access to project N / wing N** from `assertProjectAccess`.
- **Data scoping.** List queries for scoped users are filtered to `user_projects`; the demo electrical & plumbing engineers only see their wing's data.

3. Logging in & account basics

3.1 Sign in (Web Portal and Admin Console)

Both applications use the same login flow.

1. Open the app URL. If you are not signed in you are redirected to **Login**.
2. Enter your **email** and **password** and press **Sign in**.
3. On success you land on your default screen: the Web Portal Dashboard or the Admin Console Dashboard.

What happens behind the scene (data flow):

St	UI	API	Database tables	Effect
1	Submit email + password	POST /api/auth/login	users (bcrypt hash check), refresh_tokens (insert hashed token), users.last_login_at updated	Access token (short-lived) + rotating refresh token returned; JWT carries user id, roles, permission set
2	App stores tokens, calls /api/auth/me	GET /api/auth/me	users, user_roles, role_permissions, permissions, user_projects, hrms_employees (via user_id)	Full profile: roles, permission codes, project/wing access, linked employee (if any)
3	Sidebar renders	—	—	Menu items filtered by permission codes

- **Token refresh.** The API client automatically calls **POST /api/auth/refresh** when the access token expires. The old refresh token is **revoked and a new one issued** (rotation); a stolen refresh token can therefore only be used once.
- **Session revocation.** Logging out (**POST /api/auth/logout**) or an admin resetting your password revokes all of your refresh tokens immediately — every device is signed out.
- **Rate limiting.** Login endpoints are rate-limited to block credential brute force.

3.2 Forgot password

1. On the login screen click **Forgot password?**.
2. Enter your email and submit. If the account exists, the server creates a **password reset token** (single-use, time-limited) and an email link is produced by the mail pipeline configured in **backend/.env**.
3. Open the link, enter a new password (min 8 characters) on the **Reset password** screen, submit — you are back at login, and all previous sessions are revoked.

Step	UI	API	Database tables
Request	Email form	POST /api/auth/forgot-password	password_reset_tokens (new token row)
Reset	New password form	POST /api/auth/reset-password	users.password updated, password_reset_tokens consumed, refresh_tokens for the user revoked

3.3 My Profile (Web Portal)

Open the user chip (top right) → **My Profile**.

- **Left card — profile.** Shows avatar, name, email, employee code (if linked), role badges and **SUPER ADMIN** badge when applicable. You can edit **Full name** and **Phone** (saved via **PUT /api/auth/profile** → users).
- **HR & Payroll profile block** (only when your login is linked to an employee): employee code, department, effective designation, employment type, project, status. Changes made by HR on the employee record appear here automatically.
- **Project Access list:** every project (and wing, if wing-scoped) you are assigned to; Super Admins with no entries see "access to all projects".
- **Right card — change password.** Current password, new password (min 8 chars), confirm. On success you are logged out and must sign in again.

Action	API	Tables
Save name/phone	PUT /api/auth/profile	users

Action	API	Tables
Change password	POST /api/auth/change-password	users.password, all refresh_tokens revoked

4. Web Portal — complete screen & tab guide

The Web Portal is where construction work happens. The sidebar (left) is grouped into four work areas; every entry is permission-gated:

Sidebar group	Menu items
Site Operations	Dashboard · Projects · Daily Progress · Workforce & Attendance · Equipment
Project Management	Tasks & Milestones · Drawings · Inspections · Test Reports · Documents · Issues / Snags
Accounts & Sales	Materials & Inventory · BOQ · Payments · Petty Cash · CRM & Sales · HR & Payroll
Insights	Reports & Dashboards

4.0 Global chrome: top bar, search, notifications bell, user menu

- **Top bar.** Page title for the current screen, a global search field (projects, tasks, materials), the notifications bell, and the user chip (name + primary role) opening **My Profile** and **Logout**.
- **Notifications bell.** Polls `GET /api/notifications?limit=12` on open and every 60 seconds; unread count shown as a badge (capped at 99+).
 - Clicking a notification marks it read (`PUT /api/notifications/:id/read`) and **deep-links** you into the module: issues → Issues, milestones → Tasks, materials/inventory/purchase_orders → Materials (PO tab), inspections, test reports, sales, documents, drawings, hrms, petty_cash, system → home.
 - **Mark all read** via `POST /api/notifications/mark-all-read`.
 - Notification rows are tinted by type: success (green), warning (orange), error (red), info (blue).
- **User menu.** My Profile / Logout (`POST /api/auth/logout` revokes sessions).

Action	API	Tables
Open bell	<code>GET /api/notifications?limit=12</code>	notifications (per current user, unread count)
Click item	<code>PUT /api/notifications/:id/read</code>	notifications.is_read = 1
Mark all	<code>POST /api/notifications/mark-all-read</code>	notifications.is_read = 1 for all of user's rows

4.1 Dashboard

The landing page gives a cross-module snapshot of everything the organisation is doing.

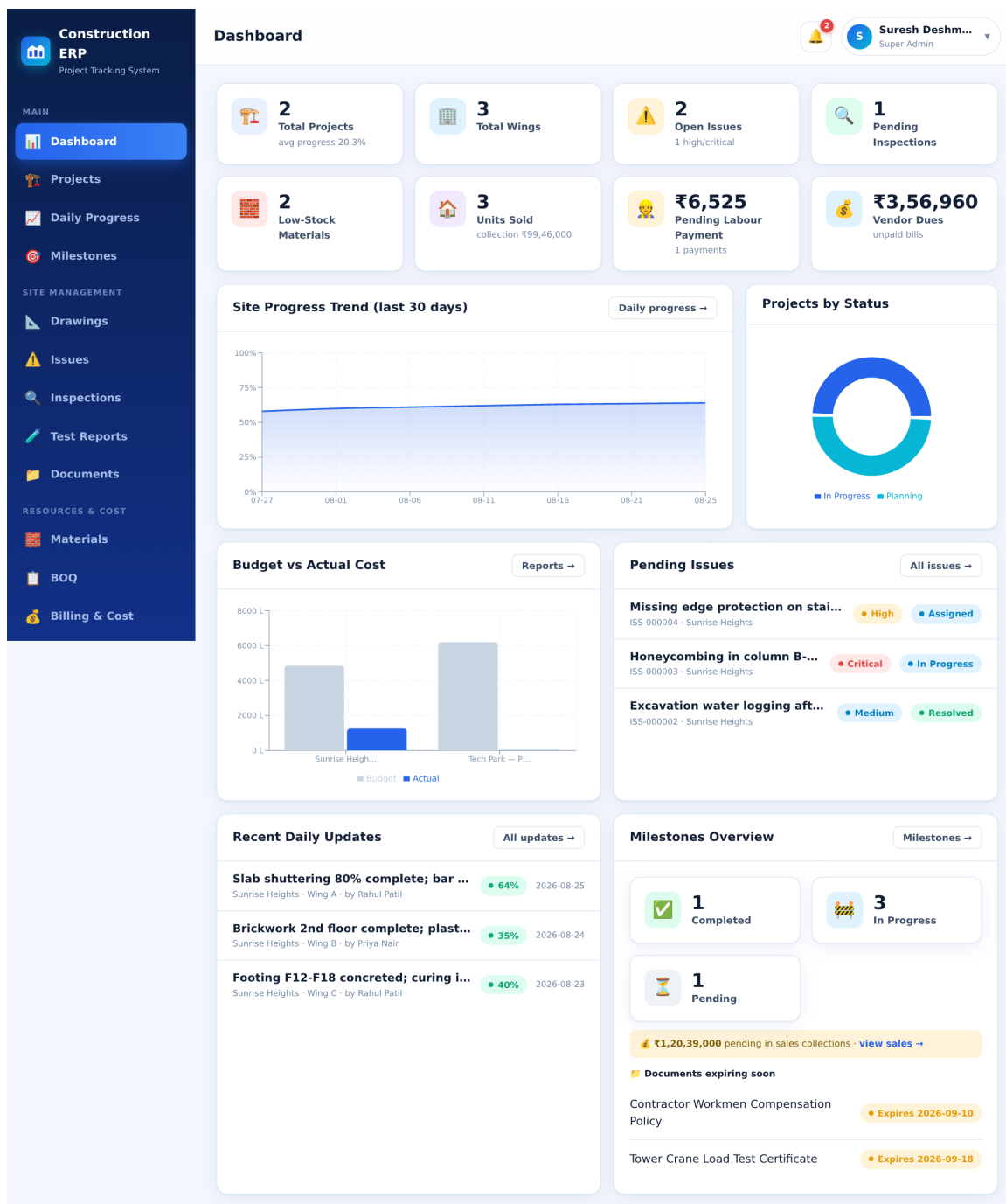
Elements

- **Hero banner** — greeting, active project count, units sold, open issues; quick buttons to Projects and a new Daily update.
- **8 KPI tiles** — Total Projects (with average progress), Total Wings, Open Issues (with high/critical count), Pending Inspections, Low-Stock Materials, Units Sold (with collections), Pending Labour Payment (amount + count), Vendor Dues (unpaid bills).
- **Charts & lists** — Site Progress Trend (avg daily progress % over last 30 days), Projects by Status (donut), Budget vs Actual Cost per project (₹ lakhs), Pending Issues list (priority + status badges), Recent Daily Updates (with %), Milestones Overview (counts by status), Sales collections callout, Documents expiring soon (top 3).
- **Module quick launch** — 10 tiles deep-linking into the core modules.

Data flow. One call powers the whole page:

UI	API	Tables read (join/aggregate)
Entire dashboard	<code>GET /api/dashboard/overview</code>	projects, wings, units, daily_progress (30-day trend), issues, inspections, materials + stock_transactions (low stock), sales + sales_payments (sold/value/pending), labour_payments (pending), cost_entries (vendor dues), milestones, project_documents (expiring)

Nothing on this page is editable — it is a read-only aggregate.



Web Portal dashboard

4.2 Projects

The register of all projects you can see (scoped users see only their assignments).

Elements

- Table columns** — Code, Project (name + city/state), Type badge, Client, Manager, Wing count, Budget, Progress bar, Status badge, Expected completion.
- Filters** — Status (planning / in_progress / on_hold / completed / cancelled), Type (residential / commercial / industrial / infrastructure / mixed / other).
- Search** — name, code, client, city.
- CRUD modal (New Project)** — Name, Code (auto-generated if blank), Type, Client/owner, Developer, City, State, PIN, Address, Project manager (user directory), Budget (₹), Status, Start / Expected / Actual completion dates, Description.

Data flow

Action	API	Tables
List + search + filter	GET /api/projects?search=&status=&project_type=&page=&limit=	projects (+ wings count, manager name join), filtered by user_projects scope
Create	POST /api/projects	projects (code auto-generated like PRJ-#### when blank), audit_logs
Edit	PUT /api/projects/:id	projects (old/new values captured in audit), audit_logs
Delete	DELETE /api/projects/:id	projects (blocked while child records exist — wings, progress, etc.), audit_logs
Export	GET /api/projects/export	same query → CSV download; audit_logs action=export

Status values: `planning` → `in_progress` → `on_hold` → `completed` | `cancelled` (managed by the manager; "cancelled" and "completed" are terminal for reporting).

Construction ERP

Project Tracking System

MAIN

Dashboard

Projects

Daily Progress

Milestones

SITE MANAGEMENT

Drawings

Issues

Inspections

Test Reports

Documents

RESOURCES & COST

Materials

BOQ

Billing & Cost

Projects

2

Suresh Deshm...

Super Admin

↓ Export CSV

+ New Project

Search

Status

Type

Search name, code, client, city...

All

All

CODE	PROJECT	TYPE	CLIENT	MANAGER	WINGS	BUDGET	PROGRESS	STATUS
PRJ - 000001	Sunrise Heights Nagpur	• Residential High-Rise	Skyline Developers LLP	Amit Verma	—	₹48,50,00,000	38%	• In Progress
PRJ - 000002	Tech Park — Phase 2 Nagpur	• Commercial	InfoSpace Realty	Amit Verma	—	₹62,00,00,000	2%	• Planning

2 records · page 1 of 1

< PrevNext >

Projects list

4.3 Project Details

Open any project row. The page has a header (name, status, type, code, client, city/state, manager, **overall progress bar** — always server-computed) and tabs:

Tab 1 — Overview. KPI tiles for wings/floors/units and budget, the progress roll-up summary, and the structural CRUD for this project: add **Wing**, add **Floor**, add **Unit** (unit number, wing, type, saleable area, price, status). These are the building blocks used by progress reports, sales and scoping.

Tab 2 — Wings (n). Wings table with per-wing progress bar, status, floor/unit counts; add/edit wing (name, code, manager, area, dates, status, progress — blank progress is auto-calculated), delete wing (with confirmation: `"Delete wing ... and all its floors/units?"`).

Tab 3 — Wing Dashboard (appears when you deep-link to a wing, e.g. from a notification or the wings tab). KPIs: wing progress %, active issues, units sold + value, collections + pending. Plus Floor Progress list (bars per floor), Recent Updates for the wing, Inspections by Status donut, Material Status mini-table (in-stock + LOW badge for that project), Active Issues table.

Tab 4 — Progress Timeline. Area chart of daily average progress % per date for the project (labour and

Tab 5 — Floors & Units. Wing filter, unit status summary badges (available / booked / sold counts), Floors table (wing, floor, progress bar, status) and Units table (unit no, wing, type, area sqft, price, status badge).

Data flow

Action	API	Tables
Load project	GET /api/projects/:id	projects + nested wings
Add wing	POST /api/wings	wings, audit_logs
Edit wing	PUT /api/wings/:id	wings (blank progress left to roll-up), audit_logs
Delete wing	DELETE /api/wings/:id	wings + cascade of its floors/units, audit_logs
Add floor	POST /api/floors	floors, audit_logs
Edit floor	PUT /api/floors/:id	floors, audit_logs
Add unit	POST /api/units	units, audit_logs
Edit / delete unit	PUT/DELETE /api/units/:id	units, audit_logs (delete blocked when the unit is sold)
Wing dashboard	GET /api/dashboard/wing/:wingId	wings, floors, daily_progress, issues, sales+sales_payments, inspections, stock_transactions+materials
Progress timeline	GET /api/progress/timeline?projectId=	daily_progress grouped by date
Floors / units lists	GET /api/floors?projectId=&wingId= · GET /api/units?projectId=&wingId=	floors · units

Roll-up reminder: whenever any **daily_progress** row for a wing changes, the server runs **recomputeProjectProgress(projectId)** inside the same transaction — wing progress = AVG(percentage) of the wing's progress rows, project **overall_progress** = average of wing progresses (capped at 100).

Construction ERP
 Project Tracking System

MAIN

- Dashboard
- Projects**
- Daily Progress
- Milestones

SITE MANAGEMENT

- Drawings
- Issues
- Inspections
- Test Reports
- Documents

RESOURCES & COST

- Materials
- BOQ
- Billing & Cost

Project Details

Suresh Deshm...
 Super Admin

Sunrise Heights
In Progress
Residential High-Rise

Overall progress
38%

PRJ-000001 · Skyline Developers LLP · Nagpur · Manager: Amit Verma

📅 Start: 2025-03-01 📅 Due: — 💰 Budget: ₹48,50,00,000

Overview
Wings (3)
 Progress Timeline
 Floors & Units

Developer	—
Address	Plot 42, Wardha Road, Nagpur
Description	—
Budget	₹48,50,00,000
Actual cost	₹10,13,960 (0.21% of budget)
Paid / Pending	₹6,57,000 / ₹3,56,960

Project detail with tabs

4.4 Daily Progress

The daily site worksheet — the heartbeat of the system. Site teams record what happened, where, with photos and GPS.

List screen

opens the coordinates on OpenStreetMap), Reported by.

- **Filters** — Project, Wing, free-text search on descriptions. Pagination (15/page).
- **Actions** — Export CSV (`progress.export`), + New Report (`progress.create`), row click → detail.

New Report modal

Field	Notes
Project / Wing / Floor	Cascading selects (wing list reloads per project, floors per wing)
Date	Report date (required)
Time	Work time
Overall progress %	Required, 0–100 — feeds the roll-up
Labour count	People on site that day
Weather	Clear / Cloudy / Rainy / Hot / Cold
Work description / Work completed today	Required / optional text
Material used, Remarks	Free text
GPS	Capture GPS button uses the browser geolocation; coordinates are stored on the report *and* attached to every photo
Site photos	Up to 8 images (jpeg/png/webp). Previews show the GPS + capture date; individual removal

Save. The form is submitted as `multipart/form-data` with a generated `client_ref` (`web-<timestamp>-<rand>`) and a `photosMeta` JSON array (per-photo latitude/longitude/captured_at/source). If the browser is offline the report is **queued in localStorage under `cerp.offlineProgress`** and the toast explains it will sync when online — the `client_ref` guarantees the server never creates it twice.

Detail modal. All fields, GPS link, photo grid (each photo shows source — app camera vs uploaded — and its coordinates), and Delete (`progress.delete`).

Data flow

Action	API	Tables	Side effects
List	GET /api/progress?projectId=&wingId=&search=&page=	daily_progress + projects, wings, floors, progress_photos counts	Scope-filtered
Create	POST /api/progress (multipart)	file_uploads (each photo), progress_photos (progress_id, file_id, lat, lng, captured_at, source, uploaded_by), daily_progress (incl. client_ref)	Progress roll-up recomputed for the project; audit_logs; duplicate guard: if client_ref already exists the existing row is returned instead of a new insert
Edit	PUT /api/progress/:id	daily_progress	Roll-up recomputed, audit_logs
Delete	DELETE /api/progress/:id	daily_progress (+ photos)	Roll-up recomputed, audit_logs
Batch sync (mobile)	POST /api/progress/sync	as create, per item	Per-item result: created / duplicate / error
Export	GET /api/progress/export	daily_progress	CSV; audit_logs action=export
Photos display	GET /api/files/:id	file_uploads	Authenticated object URL (token-bounded), served from disk

Construction ERP
Project Tracking System

MAIN

Dashboard
Projects
Daily Progress
Milestones

SITE MANAGEMENT

Drawings
Issues
Inspections
Test Reports
Documents

RESOURCES & COST

Materials
BOQ
Billing & Cost

Suresh Deshm...
Super Admin

Daily Progress

Daily Progress Reports

Export
New Report

Search

Search descriptions...

Project

All projects

Wing

All wings

DATE	WORK	PROGRESS	LABOUR	PHOTOS	GPS	BY
2026-08-25	Slab shuttering 80% complete; bar ... Sunrise Heights · Wing A · Floor 7	64%	38	2	map	Rahul Patil
2026-08-24	Brickwork 2nd floor complete; plast... Sunrise Heights · Wing B · Floor 3	35%	26	1	map	Priya Nair
2026-08-23	Footing F12-F18 concreted; curing i... Sunrise Heights · Wing C	40%	22	1	map	Rahul Patil

3 records · page 1 of 1

Prev
Next

Daily Progress

4.5 Workforce & Attendance

Three tabs: **Attendance** (default), **Workers**, **Labour Payments**.

Tab: Workers — the labour master.

- **Columns** — Code, Name, Category, Contractor, Mobile, Daily wage, OT/hr, Project, Active status.
- **Filters** — Project, Category.
- **CRUD fields** — Name, Worker code (auto if blank), Mobile, Category (from Masters → Labour Categories), Contractor (from Masters → Contractors), Daily wage (₹), Overtime rate (₹/hr), Project, Wing, Joining date, Active, ID proof.

Action	API	Tables
List/create/edit/delete	GET POST PUT DELETE /api/workers...	workers (+ category/contractor joins), audit_logs

Tab: Attendance — mark a project for one day and save all workers in one batch.

1. Pick **Project** and **Date** (default today).
2. The summary badges show counts per status (present / absent / half day / leave / overtime) plus the **pay roll** badge — the day's total wage once marked.
3. Quick-mark buttons: **All present** / **All absent**.
4. Per worker: status buttons **present** · **absent** · **half day** · **leave** · **overtime**, an **OT hrs** numeric input, and a live **Day pay** preview:
 - present or overtime → full daily wage
 - half day → 50% of daily wage
 - OT → hours × the worker's OT rate
5. **Save attendance (n)** submits every worker for that day in one call; existing rows are updated, new rows inserted.

Action	API	Tables	Side effects
--------	-----	--------	--------------

Action	API	Tables	Side effects
Load workers for project	GET /api/workers?projectId=&is_active=1	workers	—
Load existing marks	GET /api/attendance?projectId=&date=	attendance	Hydrates the mark buttons
Summary badges	GET /api/attendance/summary?projectId=&date=	attendance GROUP BY status	—
Batch save	POST /api/attendance {project_id, date, records:[{worker_id, status, overtime_hours}]}	attendance — daily_amount and overtime_amount computed server-side from the worker's wage/OT rate (same formula as the preview); upsert per worker+date+shift	audit_logs; drives Labour Payments & monthly report
Single edit	PUT /api/attendance/:id	attendance	Recomputed amounts
Export	GET /api/attendance/export?projectId=	attendance	CSV; audited
Monthly wage report	GET /api/attendance/monthly-report?projectId=&month=	attendance (days, OT, gross, paid vs pending per worker)	Read-only

Construction ERP
Project Tracking System

MAIN

- Dashboard
- Projects
- Daily Progress
- Milestones

SITE MANAGEMENT

- Drawings
- Issues
- Inspections
- Test Reports
- Documents

RESOURCES & COST

- Materials
- BOQ
- Billing & Cost

Workforce & Attendance

Attendance

Workers

Labour Payments

Daily Attendance

Export

Project

Date

Pay Roll ₹3,230

Quick mark:

All present

All absent

Save attendance (5)

CODE	WORKER	STATUS	OT HRS	DAY PAY
WKR-001	Ramesh Yadav Mason · ₹950/day	present absent half day leave overtime	0	₹950
WKR-002	Santosh Kumar Bar Bender · ₹900/day	present absent half day leave overtime	2	₹1,130
WKR-003	Prakash Jadhav Carpenter · ₹1000/day	present absent half day leave overtime	0	₹500
WKR-004	Dinesh Wagh Helper · ₹650/day	present absent half day leave overtime	0	₹650
WKR-005	Mahesh Bhoyar Helper · ₹650/day	present absent half day leave overtime	0	—
WKR-006	Anil Gawande Electrician · ₹1050/day	present absent half day leave overtime	0	₹0

Workforce — Attendance tab

Tab: Labour Payments — generate and settle worker wages from attendance.

- **Columns** — Worker (+ code), Period (start → end), Days, OT hrs, Net, Paid, Status badge, **Mark paid** action for unpaid rows.
- **Pending KPI** — sum of (net – paid) across all rows.
- **Generate Payment modal** — Worker, Period start/end, Deductions (₹), optional Pay now (₹). The server sums the worker's attendance over the period: **gross** = $\Sigma \text{daily_amount} + \Sigma \text{overtime_amount}$, **net** = **gross** – **deductions**, writes the payment row with status **pending** (or **partial/paid** if money is given at generation).
- **Mark paid** sets **paid_amount** = **net_amount**, today's date, mode cash → status **paid**.

Action	API	Tables
List	GET /api/labour-payments?projectId=	labour_payments + worker join

Action	API	Tables
Generate	POST /api/labour-payments {worker_id, project_id, period_start, period_end, deductions, paid_amount?}	attendance (read over period) → labour_payments (computed totals), audit_logs
Mark paid / edit	PUT /api/labour-payments/:id	labour_payments (paid_amount, payment_date, payment_mode, status), audit_logs

Construction ERP
Project Tracking System

MAIN

Dashboard
Projects
Daily Progress
Milestones

SITE MANAGEMENT

Drawings
Issues
Inspections
Test Reports
Documents

RESOURCES & COST

Materials
BOQ
Billing & Cost

Workforce & Attendance

2
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Attendance Workers **Labour Payments**

Labour Payments

₹6,525
Pending

+ Generate Payment

Project

All projects

WORKER	PERIOD	DAYS	OT HRS	NET	PAID	STATUS
Ramesh Yadav WKRR-001	2026-08-01 → 2026-08-15	—	—	₹11,400	₹11,400	Paid
Santosh Kumar WKRR-002	2026-08-01 → 2026-08-15	—	—	₹12,525	₹6,000	Partial Mark paid

Workforce — Labour Payments tab

4.6 Equipment

Track heavy machinery and site equipment: fleet status, daily hour logs, fuel, and monthly hire billing.

Summary tiles — Total fleet (with deployed count), Available, Deployed, Maintenance, Breakdown, **Hours this month** (running hours) with fuel used in litres. From GET /api/equipment/summary.

Fleet table

- **Columns** — Code, Equipment, Type (excavator, crane, concrete mixer, tower crane, bulldozer, loader, generator, compactor, dumper, scaffolding, pump, other), Ownership (owned / rented / contractor supplied), Project, Hourly rate, Status badge (available / deployed / maintenance / breakdown / idle), Last log date.
- **Filters** — Search (code, name, operator), Status, Type, Ownership.
- **Add / Edit equipment fields** — Code (auto EQ-... if blank), Name, Type, Ownership, Project ID, Wing ID, Hourly / Daily / Monthly rate, Capacity, Registration no / asset tag, Operator, Status, Deployed on, Active, Remarks. Delete is allowed only when the unit has **no logs** (history is preserved).

Equipment detail modal (row click or Open)

- **Profile card** — status, ownership, rates, registration, operator.
- **Daily log form** — Date, Status after, Deployed hrs, Running hrs, Idle hrs, Breakdown hrs, Fuel (L), Work done, Remarks. Saving inserts an **equipment_logs** row **and flips equipment.status to "status after"** — so the fleet board always reflects the latest log.
- **Log history table** — date, hours, fuel, status after, remarks.

CRUD fields — Project, Name, Start date, Target date, Progress % (entering 100 auto-completes the milestone — status flips to completed), Status, Responsible person (user directory), Completion date, Description, Remarks.

- Overdue milestones (past target, not completed) surface as **delayed** in the dashboard overview.

Action	API	Tables	Side effects
List/create/edit/delete	GET POST PUT DELETE /api/milestones...	milestones (+ project, responsible user joins), audit_logs	Delay detection → notifications event milestone_delayed to users with milestones.approve

Construction ERP

Project Tracking System

MAIN

Dashboard

Projects

Daily Progress

Milestones

SITE MANAGEMENT

Drawings

Issues

Inspections

Test Reports

Documents

RESOURCES & COST

Materials

BOQ

Billing & Cost

Milestones

2

S

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+ New Milestone

Search

Project

Status

Search...

All

All

MILESTONE	START	TARGET	PROGRESS	STATUS	RESPONSIBLE	
Foundation & Plinth Completion Sunrise Heights - Wing A	2025-05-01	2025-08-31	100%	Completed	Rahul Patil	<button>Edit</button> <button>Delete</button>
Superstructure — Slabs up to 7th Sunrise Heights - Wing A	2025-09-01	2026-02-28	62%	In Progress	Rahul Patil	<button>Edit</button> <button>Delete</button>
Brickwork & Plaster Sunrise Heights - Wing B	2025-11-01	2026-05-31	35%	In Progress	Priya Nair	<button>Edit</button> <button>Delete</button>
Foundation Completion Sunrise Heights - Wing C	2026-01-01	2026-04-30	40%	In Progress	Rahul Patil	<button>Edit</button> <button>Delete</button>
MEP First Fix Sunrise Heights	2026-06-01	2026-10-31	0%	Pending	Vikram Joshi	<button>Edit</button> <button>Delete</button>

5 records · page 1 of 1

Prev

Next

Milestones

4.8 Drawings

Version-controlled drawing control with revision approval.

List — Drawing No., Title (+ project · wing), Category badge (architectural / structural / electrical / plumbing / hvac / fire_fighting / interior / other), current **Rev** badge, Approval status, Uploaded by. Filters: search, project, category.

Upload Drawing — Project, Wing (optional), Drawing number (required), Category, Title (required), First revision no (default R0), Revision date, File (PDF/image), Remarks. The file is stored in **file_uploads** and the first row of **drawing_revisions** is created.

Drawing detail modal

- Header: project, category, status, latest revision.
- Revision History** table — Rev, Date, status, uploaded by, remarks, preview/download buttons.
- Add revision** — revision no, date, remarks, file → new revision row with status **pending**.
- Approve / Reject revision** (when **drawings.approve**) — on approval the drawing's **approval_status** becomes **approved** and the new file is the current one; on rejection it goes to **rejected**.

Action	API	Tables	Side effects
List	GET /api/drawings?page=&search=&projectId=&category=	drawings (+ latest revision), file_uploads	—
Upload	POST /api/drawings (multipart)	file_uploads, drawings, drawing_revisions (R0), audit_logs	notifications event drawing_revision to drawing approvers
Detail	GET /api/drawings/:id	drawings + drawing_revisions + files	—

Action	API	Tables	Side effects
Add revision	POST /api/drawings/:id/revisions (multipart)	file_uploads, drawing_revisions (status pending), audit_logs	Notification to approvers
Approve/reject	PUT /api/drawings/revisions/:revisionId/status	drawing_revisions, drawings.approval_status + latest revision updated	audit_logs, notification
Preview / download	GET /api/files/:id	file_uploads	Authenticated object URL


Construction ERP
Project Tracking System

MAIN

Dashboard
Projects
Daily Progress
Milestones

SITE MANAGEMENT

Drawings
Issues
Inspections
Test Reports
Documents

RESOURCES & COST

Materials
BOQ
Billing & Cost

Drawings




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Drawings & Document Control

+ Upload Drawing

Search

Drawing number or title...

Project

All

Category

All

DRAWING NO.	TITLE	CATEGORY	REV	STATUS	UPLOADED BY
SH-A-ST-101	Foundation Layout Plan — Wing A Sunrise Heights · Wing A	Structural	R2	Approved	Vikram Joshi
SH-A-ST-201	Typical Floor Framing — Wing A Sunrise Heights · Wing A	Structural	R1	Approved	Vikram Joshi
SH-B-AR-102	Unit Plan Types A/B — Wing B Sunrise Heights · Wing B	Architectural	R2	Pending Approval	Vikram Joshi

3 records · page 1 of 1

Prev
Next

Drawings

4.9 Inspections

On-site inspection checklists (safety, pre-pour, finishing, etc.).

List — No. (INS-...), Type badge, Project (+ wing · location), Date, Inspector, GPS link, Status badge (pending / passed / failed / reinspection_required / closed). Filters: project, status. **Close** action for pending/failed/reinspection rows (requires **inspections.approve**).

New Inspection modal

- Inspection type (from Masters → Inspection Types; the type carries a **checklist template** of items).
- Project, Wing, Location, Date/time, Inspector (user directory).
- Checklist** — one row per item with result **pass / fail / N-A**; add or remove rows.
- Capture GPS** (browser geolocation) — stored with the inspection.
- Save → status **pending**; the result status is derived (any fail → **failed**, else **passed**).

Action	API	Tables	Side effects
List	GET /api/inspections?page=&projectId=&status=	inspections (+ type, project, inspector)	—
Create	POST /api/inspections { ...form, items:[{checklist_item, result}] }	inspections, inspection_items (one row per checklist line), audit_logs	Failed result → notifications event inspection_failed; type checklist loaded from inspection_types.checklist_template
Detail	GET /api/inspections/:id	inspections + inspection_items	—

Action	API	Tables	Side effects
Update / close	PUT /api/inspections/:id	inspections, audit_logs	—


Construction ERP
Project Tracking System

MAIN

Dashboard
Projects
Daily Progress
Milestones

SITE MANAGEMENT

Drawings
Issues
Inspections
Test Reports
Documents

RESOURCES & COST

Materials
BOQ
Billing & Cost

Inspections




Inspection Reports

+ New Inspection

Project: All Status: All

NO.	TYPE	PROJECT	DATE	INSPECTOR	GPS	STATUS
INSP-000002	Pre-Slab Pour Checklist	Sunrise Heights Wing A · Wing A 6th slab	2026-08-23	Neha Desai		Passed
INSP-000001	Scaffolding Safety Inspection	Sunrise Heights Wing B · Wing B external scaffold	2026-08-21	Sunil Pawar		Reinspection Required

Inspections

4.10 Test Reports (quality)

Material laboratory test register with an approve/reject workflow.

List — Test No., Test (type), Project (+ wing), Material, Test date, Laboratory, **Result** (pending / pass / fail / inconclusive), status, ↓ **PDF** download when a file is attached. Filters: result, test type.

New Test Report fields — Project, Wing, Test type (from Masters → Test Types), Material (optional), Sample date, Test date, Laboratory, Standard/spec, Result (default pending), Test result / values (multi-line), Remarks, report file upload.

Workflow — when a report is in status **submitted**, approvers (`test_reports.approve`) get inline **Approve / Reject** buttons.

Action	API	Tables	Side effects
List/create/edit/delete	GET POST PUT DELETE /api/test-reports...	test_reports (+ type, material joins), file_uploads when a PDF is attached, audit_logs	result_status = fail → notifications event test_failed to quality approvers
Approve/reject	PUT /api/test-reports/:id/status	test_reports.status (submitted → approved/rejected) + approved_by/at	audit_logs, notification
PDF download	GET /api/files/:id	file_uploads	Authenticated object URL

Construction ERP

Project Tracking System

MAIN

Dashboard

Projects

Daily Progress

Milestones

SITE MANAGEMENT

Drawings

Issues

Inspections

Test Reports

Documents

RESOURCES & COST

Materials

BOQ

Billing & Cost

Test Reports

+ New Test Report

Search

Result

Test type

Search...

All

All

TEST NO.	TEST	PROJECT	MATERIAL	TEST DATE	LABORATORY	RESULT	STATUS	REPORT
TR-000003	Concrete Cube Compressive Strength	Sunrise Heights Wing A	M30 Design Mix	2026-08-25	SGBMaterials Lab, Nagpur	Pass	Approved	PDF Edit
TR-000002	Steel Tensile Strength	Sunrise Heights	TMT Steel Fe500 12mm	2026-08-21	SGS Nagpur	Pass	Approved	PDF Edit
TR-000001	Silt Content in Sand	Sunrise Heights	River Sand (Washed)	2026-08-13	In-house	Fail	Rejected	— Edit

3 records · page 1 of 1

Test Reports

4.11 Documents

Project document vault with expiry tracking.

- **Categories** — agreement, certificate, approval, noc, project image, demo image, site photo, client document, government document, other.
- **List columns** — Title, Category, Document no, Version, Expiry (highlighted **amber when expiring within 30 days**), File name, Uploaded by. Filters: category, project, search.
- **Upload** — Project, Wing (optional), Title (required), Category, Document number, Version, Expiry date, Description, File (PDF / image / Office docs).
- **Actions** — ↓ Download (authenticated file URL), Delete (with confirmation).
- Documents near expiry appear on the **Dashboard** ("Documents expiring soon") and raise the **document_expiry** notification event.

Action	API	Tables	Side effect
List	GET /api/documents?limit=&category=&projectId=&search=	project_documents (+ files, uploader)	—
Upload	POST /api/documents (multipart)	file_uploads, project_documents, audit_logs	—
Download / delete	GET /api/files/:id · DELETE /api/documents/:id	file_uploads · project_documents	Audited

Construction ERP
Project Tracking System

MAIN

Dashboard
Projects
Daily Progress
Milestones

SITE MANAGEMENT

Drawings
Issues
Inspections
Test Reports

Documents

RESOURCES & COST

Materials
BOQ
Billing & Cost

Documents

2
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Project Documents
+ Upload Document

Search
Title, number, description...

Project
All

Category
All

DOCUMENT	CATEGORY	DOC NO	VER	EXPIRY	FILE	UPLOADED BY	
Commencement Certificate — Wing A-C Sunrise Heights	Approval	—	—	—	cc-wings.pdf	Amit Verma	Download Delete
Contractor Workmen Compensation Policy Sunrise Heights	Insurance	—	—	2026-09-10	wc-policy.pdf	Anita Rao	Download Delete
Tower Crane Load Test Certificate Sunrise Heights	Equipment	—	—	2026-09-18	crane-load-test.pdf	Sunil Pawar	Download Delete
Fire NOC — Provisional Sunrise Heights	Approval	—	—	2026-12-31	fire-noc.pdf	Amit Verma	Download Delete

Documents

4.12 Issues / Snags

Field issues with priority, assignment, due dates, photos and a discussion thread.

List — No. (ISS-...), Issue (title + project · wing · location), Category, Priority badge (low / medium / high / critical), Status badge, Assigned to, Due (red when overdue and not resolved/closed). Filters: project, status, priority.

Raise Issue modal — Project, Wing, Title (required), Category (from Masters → Issue Categories), Priority, Assign to (user directory), Due date, Location, Description, optional photos (multipart).

Issue detail modal

- Full header: project/wing/location, priority + category, raised by (+ GPS link), due date, description, photos.
- Status & assignment editor** (visible to `issues.edit` holders or the assignee): status select and assignee select → Save.
- Discussion** — chronological comment thread with author + timestamp; add comments inline (Enter or Send).

Status workflow: open → assigned → in_progress → resolved → closed, with **reopen** from resolved/closed back to in_progress.

Action	API	Tables	Side effects
List	GET /api/issues?page=&projectId=&status=&priority=	issues (+ category, assignee, project)	—
Raise	POST /api/issues (multipart)	issues (number auto ISS-####), file_uploads+issue photos when attached, audit_logs	notifications issue_created to watchers; issue_assigned to the assignee when set
Detail	GET /api/issues/:id	issues + comments + photos	—
Update status/assignee	PUT /api/issues/:id	issues, audit_logs	notifications issue_updated to affected users
Comment	POST /api/issues/:id/comments	issue_comments, audit_logs	—
Overdue sweep	server-side on reads	issues	notifications issue_overdue

Page 24 / 62

Construction ERP

Project Tracking System

MAIN

Dashboard

Projects

Daily Progress

Milestones

SITE MANAGEMENT

Drawings

Issues

Inspections

Test Reports

Documents

RESOURCES & COST

Materials

BOQ

Billing & Cost

Issues

Site Issues

+ Raise Issue

Project

Status

Priority

All

All

All

NO.	ISSUE	CATEGORY	PRIORITY	STATUS	ASSIGNED TO	DUE
ISS-000004	Missing edge protection on stairwell opening Sunrise Heights · Wing A · Wing A, 5th floor stairwell	Safety	High	Assigned	Rahul Patil	2026-08-26
ISS-000003	Honeycombing in column B-G2 Sunrise Heights · Wing B · Wing B GF	Quality	Critical	In Progress	Rahul Patil	2026-08-24
ISS-000002	Excavation water logging after rain Sunrise Heights · Wing C · Wing C excavation	Schedule	Medium	Resolved	Sanjay Kale	2026-08-20
ISS-000001	Debris stacking near hoist Sunrise Heights · Wing A	Housekeeping	Low	Closed	—	—

4 records · page 1 of 1

< Prev

Next >

Issues / Snags

4.13 Materials & Inventory

Seven tabs covering the complete material lifecycle — the largest module in the portal:

Tabs: Materials · Stock & Alerts · Requirements · Purchase Orders · Receipts (GRN) · Consumption · Suppliers

The tab is deep-linkable via ?tab=pos etc. (notifications to POs land you on the right tab).

Tab 1 — Materials (master)

- **Columns** — Code, Name, Category, Unit, Min Stock, Active.
- **Filter** — Category. **Search** — name/code.
- **CRUD fields** — Name, Code (required), Category (from Masters → Material Categories), Unit (bags / kg / ton / cft / cum / sqft / sqm / nos / meter / litre / month), **Min stock level** (the alert threshold), Active, Description.
- Export CSV available.

Action	API	Tables
List/create/edit/delete	GET POST PUT DELETE /api/materials...	materials, audit_logs
Categories	GET POST PUT DELETE /api/materials/categories...	material_categories (managed from Admin Masters)

Construction ERP
Project Tracking System

MAIN

Dashboard
Projects
Daily Progress
Milestones

SITE MANAGEMENT

Drawings
Issues
Inspections
Test Reports
Documents

RESOURCES & COST

Materials
BOQ
Billing & Cost

Materials & Inventory

Materials
Stock & Alerts
Requirements
Purchase Orders
Receipts (GRN)
Consumption
Suppliers

Materials

Search

Search...

Category

All

CODE	MATERIAL	CATEGORY	UNIT	MIN STOCK	STATUS	
STL-001	TMT Steel Fe500 12mm	Steel	kg	2,000	Active	Edit Delete
CEM-001	Cement OPC 53 Grade	Cement	bag	150	Active	Edit Delete
SAND-001	River Sand (Washed)	Aggregates	cft	500	Active	Edit Delete
AGG-001	Aggregate 20mm	Aggregates	cft	400	Active	Edit Delete
BLK-001	AAC Blocks 600x200x150	Masonry	nos	3,000	Active	Edit Delete
PNT-001	Exterior Emulsion (White)	Finishing	ltr	100	Active	Edit Delete

6 records · page 1 of 1

Prev
Next

Materials master tab

Tab 2 — Stock & Alerts

The live stock position, **computed from the transactions ledger** — there is no hand-edited stock number.

- **Columns** — Material (+ code), Project, Received, Consumed, Damaged, Returned, **Current Stock** (qty + unit), **Alert** (LOW STOCK badge when `current_stock <= min_stock_level` and the level is set).
- **Filters** — search, project. **Refresh** button.
- **Transactions** toggle reveals the recent **stock ledger**: date, material, type (opening / receipt / consumption / return — receipts & openings green, others red), quantity (negative in red), reference module + id, by whom.

UI	API	Tables
Stock position	GET /api/materials/stock?search=&projectId=	stock_transactions aggregated per material/project (received = Σ + receipts, consumed = Σ -, damaged, returned) + materials for min level
Ledger	GET /api/materials/stock/transactions?projectId=	stock_transactions (newest first)
Low-stock alerts feed	GET /api/materials/stock/low-stock-alerts	same aggregation, filtered to <code>low_stock = 1</code>

Page 26 / 62

Construction ERP
Project Tracking System

MAIN

Dashboard
Projects
Daily Progress
Milestones

SITE MANAGEMENT

Drawings
Issues
Inspections
Test Reports
Documents

RESOURCES & COST

Materials
BOQ
Billing & Cost

Materials & Inventory

2

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Materials
Stock & Alerts
Requirements
Purchase Orders
Receipts (GRN)
Consumption
Suppliers

Stock Position

Transactions
 Refresh

Search

Material name or code...

Project

All

MATERIAL	PROJECT	RECEIVED	CONSUMED	DAMAGED	RETURNED	CURRENT STOCK	ALERT
TMT Steel Fe500 12mm STL-001	Sunrise Heights	29,000	21,200	0	0	7,800 kg	
Cement OPC 53 Grade CEM-001	Sunrise Heights	2,600	2,510	5	0	85 bag	
River Sand (Washed) SAND-001	Sunrise Heights	3,400	2,210	0	0	1,190 cft	
Aggregate 20mm AGG-001	Sunrise Heights	2,600	1,710	0	0	890 cft	
AAC Blocks 600x200x150 BLK-001	Sunrise Heights	26,000	23,300	0	0	2,700 nos	

Stock & Alerts tab

Tab 3 — Requirements

Raise a material need; store approves; procurement follows.

- **Columns** — Req No. (REQ-...), Material, Project (+ wing), Qty + unit, Needed by, Requested by, Status.
- **Filter** — Status: pending → approved | rejected → po_created → fulfilled.
- **New Requirement fields** — Project, Wing, Material, Required qty, Unit, Required date, Remarks.
- **Approve / Reject** buttons on pending rows (`materials.approve`).

Action	API	Tables	Side effects
List/create/edit/delete	GET POST PUT DELETE /api/materials/requirements...	material_requirements, audit_logs	—
Approve / Reject	PUT /api/materials/requirements/:id/status	material_requirements (status, approved_by, approved_at)	notifications event po_approval (requirement decision) to the requester

Page 27 / 62

Construction ERP
Project Tracking System

MAIN

Dashboard
Projects
Daily Progress
Milestones

SITE MANAGEMENT

Drawings
Issues
Inspections
Test Reports
Documents

RESOURCES & COST

Materials
BOQ
Billing & Cost

Materials & Inventory

S
Suresh Deshm...
Super Admin

Materials
Stock & Alerts
Requirements
Purchase Orders
Receipts (GRN)
Consumption
Suppliers

Material Requirements

+ New

Search

Search...

Status

All

REQ NO.	MATERIAL	PROJECT	QTY	NEEDED BY	REQUESTED BY	STATUS	
	TMT Steel Fe500 12mm	Sunrise Heights Wing A	— kg	—	Rahul Patil	Approved	Edit Delete
	Cement OPC 53 Grade	Sunrise Heights	— bag	—	Priya Nair	Pending	Approve Reject Edit Delete
	Aggregate 20mm	Sunrise Heights Wing C	— cft	—	Rahul Patil	Pending	Approve Reject Edit Delete

3 records · page 1 of 1

< Prev

Next >

Requirements tab

Tab 4 — Purchase Orders

- **Columns** — PO No. (PO-...), Supplier, Project, Date, Expected delivery, Grand total, Status.
- **Filter** — Status: draft → pending_approval → approved → sent → partially_received → received, plus cancelled.
- **Row actions** — **Submit** (draft → pending_approval), **Approve** (→ approved, materials.approve), **Mark sent** (→ sent).
- **New PO modal** — Project, Supplier, PO date, Expected delivery, Discount (₹), Remarks + **items grid**: material (unit auto-filled), qty, unit, rate ₹, tax % (default 18) per line; **Subtotal / Tax / Grand total computed live** (subtotal = $\sum \text{qty} \times \text{rate}$; tax per line; grand = subtotal + tax – discount).
- **PO detail modal** — header (supplier, project, status, total incl. tax), items table (ordered vs received qty, rate, amount), and — when status is approved/sent/partially_received — the **Record Receipt (GRN)** panel: per item "Received" and "Damaged" inputs showing pending quantity.

Action	API	Tables	Side effects
List	GET /api/materials/purchase-orders?page=&status=	purchase_orders + supplier/project, item sums	—
Create	POST /api/materials/purchase-orders {...header, items:[{material_id, quantity, unit, rate, tax_percent}]}	purchase_orders (totals computed server-side: subtotal, tax, grand), purchase_order_items, audit_logs	—
Detail	GET /api/materials/purchase-orders/:id	PO + items (with running received_qty)	—
Status changes	PUT /api/materials/purchase-orders/:id/status	purchase_orders (+ approved_by/at)	Approve → notification to creator (po_approval)

Page 28 / 62

Construction ERP
Project Tracking System

MAIN

Dashboard
Projects
Daily Progress
Milestones

SITE MANAGEMENT

Drawings
Issues
Inspections
Test Reports
Documents

RESOURCES & COST

Materials
BOQ
Billing & Cost

Materials & Inventory

S
Suresh Deshm...
Super Admin

Materials
Stock & Alerts
Requirements
Purchase Orders
Receipts (GRN)
Consumption
Suppliers

Purchase Orders

+ New PO

Status

All

PO NO.	SUPPLIER	PROJECT	DATE	EXPECTED	TOTAL	STATUS
PO-000001	Mahalaxmi Steel Traders	Sunrise Heights	—	—	₹4,44,860	● Partially Received
PO-000002	UltraBuild Cement Distributors	Sunrise Heights	—	—	₹1,60,650	● Sent
PO-000003	Vidarbha Sand Suppliers	Sunrise Heights	—	—	₹1,00,800	● Completed

3 records · page 1 of 1

< Prev
Next >

Purchase Orders tab

Tab 5 — Receipts (GRN)

- **Columns** — GRN No., Date, Supplier, Project, PO Ref, Challan. Row click → detail with per-item **Received / Damaged / Accepted** (accepted = received – damaged).
- Receipts are normally recorded from the PO detail (above); the tab is the register.

Recording a GRN (data flow)

Step	API	Tables written	Side effects
Save GRN	POST /api/materials/receipts {po_id, project_id, receipt_date, items:[{po_item_id, received_qty, damaged_qty}]}	material_receipts (GRN-...), material_receipt_items (accepted_qty computed)	stock_transactions +receipt per item (accepted qty); damaged qty tracked separately; purchase_order_items.received_qty increased; PO auto-moves to partially_received or received; notifications material_received to store approvers; if stock falls to/below min level after consumption elsewhere, material_shortage alert; audit_logs

Tab 6 — Consumption

Record material used on site.

- **Columns** — Date, Material, Project, Qty + unit, Location, Wing, By.
- **Fields** — Project, Wing, Material, Quantity, Unit, Date, Location, Remarks.
- The server performs a **live stock check**: it reads current stock from the ledger and inserts a **negative stock_transactions** row; consuming more than available is rejected.

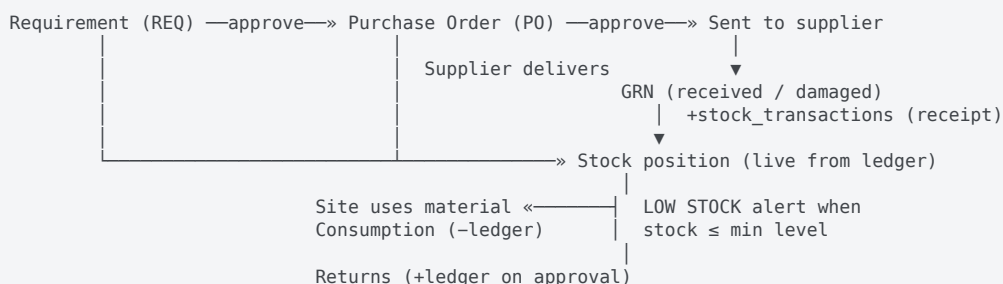
Action	API	Tables	Side effects
List/create/edit/delete	GET POST PUT DELETE /api/materials/consumption...	material_consumption + stock_transactions (–qty)	audit_logs; low-stock alert event when crossing the min level
Returns	POST /api/materials/returns · PUT /api/materials/returns/:id/approve	material_returns + stock_transactions (+qty on approval)	Approved returns add stock back

Tab 7 — Suppliers

- **Columns** — Name, Code, Contact, Phone, GST No, City.
- **Fields** — Name, Code, Contact person, Phone, Email, GST number, Address, City, State, PIN.

Action	API	Tables
CRUD	GET POST PUT DELETE /api/materials/suppliers...	suppliers, audit_logs

The complete material cycle (summary)



4.14 BOQ (Bill of Quantities)

Estimated cost baseline per project (optionally per wing), compared against actuals.

BOQ List

- **Columns** — BOQ No. (BOQ-...), Title (+ project · wing), Estimated, Actual, **Variance** (red if over budget, green if under), Status.
- **Filter** — Status: draft / active / completed / cancelled.
- **New BOQ fields** — Project, Wing (optional), Title, Tax % (default 18), Discount (₹), Status, Description.
- **↑ Import items** — paste CSV rows `item_code`, `description`, `unit`, `estimated_qty`, `rate`, `actual_qty` (first line = header); a **template** can be downloaded from `/api/boq/template-csv`. Import goes through **POST** `/api/boq/:id/import-json` (rows are parsed in the UI and posted as JSON).
- Export CSV per BOQ and for the whole list.

BOQ Detail (/boq/:id)

- **KPI band** — Estimated, Actual, Variance (₹ + %), Tax (at BOQ tax %), Grand Total.
- **Items table** — Code, Description (+ category), Unit, Est. Qty, Rate, Est. Amt, Act. Qty, Act. Amt, **Var %** (per item), Edit.
- **Add / Edit item modal** — Item code, Category (from Masters → BOQ Categories), Description, Unit, Rate, Estimated qty, Actual qty, live Estimated amount / Actual amount / Variance, Remarks.

Calculation rules (server-side): item est. amt = est. qty × rate; act. amt = act. qty × rate; BOQ totals roll up items; tax and discount applied at header level; variance % = (actual – estimated) / estimated.

Action	API	Tables
List	GET /api/boq?page=&status=&search=	boq + item total aggregation
Create	POST /api/boq	boq, audit_logs
Detail	GET /api/boq/:id	boq + boq_items + computed totals
Save item (new or edit)	PUT /api/boq/:id (whole items array) or PUT /api/boq/items/:itemId	boq_items, boq (totals recomputed), audit_logs
Import	POST /api/boq/:id/import-json	boq_items (bulk), totals recomputed
Export / template	GET /api/boq/export · GET /api/boq/:id/export · GET /api/boq/template-csv	CSV generation, audited

Construction ERP

Project Tracking System

MAIN

Dashboard

Projects

Daily Progress

Milestones

SITE MANAGEMENT

Drawings

Issues

Inspections

Test Reports

Documents

RESOURCES & COST

Materials

BOQ

Billing & Cost

Bill of Quantities

2

S

Suresh Deshm...

Super Admin

Bill of Quantities

Export CSV

New BOQ

Search

Search...

Status

All

BOQ NO.	TITLE	ESTIMATED	ACTUAL	VARIANCE	STATUS
BOQ-000001	Sunrise Heights — Civil Works BOQ Sunrise Heights	₹1,84,76,250	₹1,16,31,200	₹0	Active

1 record · page 1 of 1

PrevNext

BOQ list

Construction ERP

Project Tracking System

MAIN

Dashboard

Projects

Daily Progress

Milestones

SITE MANAGEMENT

Drawings

Issues

Inspections

Test Reports

Documents

RESOURCES & COST

Materials

BOQ

Billing & Cost

Construction ERP

2

S

Suresh Deshm...

Super Admin

Sunrise Heights — Civil Works BOQ

Active

BOQ-000001 · Sunrise Heights

Export CSV

Add item

Estimated

Actual

Variance

Tax (%)

Grand Total

BOQ Items (4)

CODE	DESCRIPTION	UNIT	EST. QTY	RATE	EST. AMT	ACT. QTY	ACT. AMT	VAR %	
CW-01	M30 RCC for slabs & columns (supply + pour) Concrete	cum	1,850	₹7,150	₹1,32,27,500	812	₹58,05,800	—	Edit
CW-02	TMT Fe500 reinforcement — cut, bend, fix Steel	kg	42,000	₹68	₹28,56,000	21,200	₹14,41,600	—	Edit
CW-03	AAC block masonry 150mm thick Masonry	sqm	4,800	₹320	₹15,36,000	1,260	₹4,03,200	—	Edit
CW-04	12mm cement plaster internal, machine finish Plaster	sqm	9,600	₹185	₹8,55,000	1,986	₹6,67,600	—	Edit

BOQ detail

Page 31 / 62

4.15 Payments (Billing & cost entries)

Record project costs against vendors and track what has been paid.

KPI band — Total Cost, Paid, Pending Payments, plus a **cost breakdown by project** stacked bar (Consumables / Equipment / Labour, ₹ lakhs) from **GET /api/billing/summary**.

Cost entries table

- **Columns** — Date, Type badge (consumable / non-consumable / labour / other), Description (+ project · wing), Category, Qty, Total, Paid, **Pending** (red when > 0), Payment status (unpaid / partial / paid).
- **Filters** — Type, Payment status, Project.
- **New Cost Entry fields** — Project, Wing, Type, Category (Cement, Bricks, Sand, Steel, Aggregate, Tiles, Pipes, Electrical, Paint, Other, Machinery, Equipment, Crane charges, Excavator charges, Vehicle charges, Rental equipment), Description, Date, Bill/invoice no, Bill date, Quantity, Unit, Rate, **Tax/GST % (default 18)**, Paid amount, Vendor/supplier (from Materials → Suppliers), Remarks. **total_amount** = qty × rate × (1 + tax) is computed server-side when quantity/rate are given; **pending_amount** = total – paid.

Action	API	Tables
List + filters	GET /api/billing/cost-entries?entry_type=&payment_status=&projectId=	cost_entries + project/supplier joins, scope-filtered
Create / edit / delete	POST PUT DELETE /api/billing/cost-entries...	cost_entries, audit_logs (payment_status derived from paid vs total)
Summary	GET /api/billing/summary · GET /api/billing/summary/by-wing	cost_entries aggregated per project (and per wing)
Export	GET /api/billing/cost-entries/export	CSV, audited

These entries feed the **Dashboard vendor dues** KPI, the **Budget vs Actual** chart/report, and the **pending-collections/project-cost** reports.

Construction ERP
Project Tracking System

MAIN

Dashboard
Projects
Daily Progress
Milestones

SITE MANAGEMENT

Drawings
Issues
Inspections
Test Reports
Documents

RESOURCES & COST

Materials
BOQ
Billing & Cost

Billing & Cost

₹58,73,960
Total Cost

₹30,57,000
Paid

₹28,16,960
Pending Payments

Cost breakdown by project (₹ Lakhs)

Cost / Billing Entries

Export CSV
+ New Cost Entry

Search
Type
Payment
Project

DATE	TYPE	DESCRIPTION	CATEGORY	QTY	TOTAL	PAID	PENDING	PAYMENT	
2026-08-22	Consumable	TMT steel — PO-000001 Sunrise Heights	material	—	₹4,44,860	₹2,00,000	₹2,44,860	Partial	Edit
2026-08-20	Labour	Shuttering gang — weekly bill Sunrise Heights · Wing A	labour	—	₹1,85,000	₹1,85,000	—	Paid	Edit
2026-08-18	Non Consumable	Tower crane rental — Aug Sunrise Heights	equipment	—	₹1,12,100	₹0	₹1,12,100	Unpaid	Edit
2026-08-14	Consumable	Cement 500 bags Sunrise Heights · Wing B	material	—	₹2,72,000	₹2,72,000	—	Paid	Edit

4 records · page 1 of 1
Prev
Next

Payments / billing

4.16 Petty Cash

Track site cash: top-ups (cash in), expenses, replenishments.

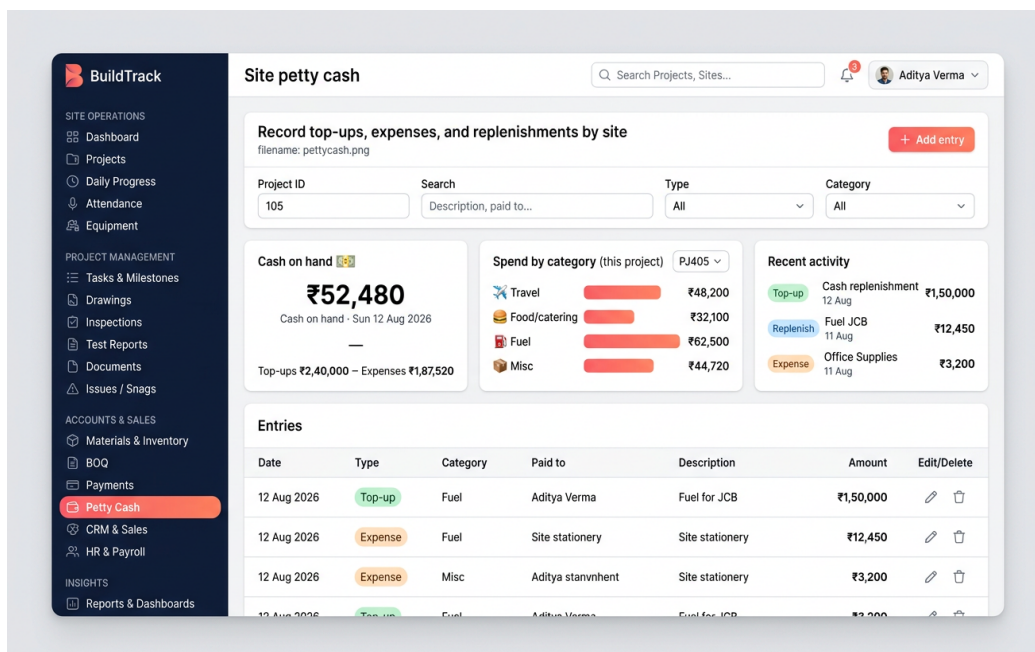
When a project is selected the summary appears:

- **Cash on hand** — big balance figure: **top-ups + replenishments – expenses**.
- **Spend by category** — horizontal bars per category (Travel / conveyance, Food / catering, Fuel & lubricants, Labour incentives, Tools & consumables, Office & admin, Misc).
- **Recent activity** — last 5 entries with type badge (topup green, replenish blue, expense orange), paid-to, amount.

Entries table — Date, Type, Category, Paid to, Description, Amount, Project. Filters: project, search, type, category. Actions: Edit / Delete (per permission), ↓ **Export**, + **Add entry**.

New entry fields — Project (required; pre-filled from the filter), Txn type (**topup** / **expense** / **replenish**), Date, Amount, Category, Paid to, Received by, Description, Remarks.

Action	API	Tables	Side effects
List	GET /api/petty-cash?projectId=&search=&txn_type=&category=&page=	petty_cash_entries + project join, scope-filtered	—
Summary	GET /api/petty-cash/summary?projectId=	petty_cash_entries aggregated (balance, by category, recent)	—
Create / edit / delete	POST PUT DELETE /api/petty-cash...	petty_cash_entries, audit_logs	—
Export	GET /api/petty-cash/export?... (honours current filters)	CSV, audited	—



Petty Cash

4.17 CRM & Sales

Sell the units that were defined under Projects → Floors & Units, and chase collections.

KPI band — Units Sold (with available count), Sales Value, Collected, **Pending Collections**, and a **Sales register** ↓ **CSV** export card. Below, a unit-availability funnel (available / booked / sold counts) from **GET** `/api/sales/unit-availability`.

Unit Sales table

- **Columns** — Unit (number + wing · floor · type), Customer (+ phone), Sale date, Amount (+ incl. GST note), Received, **Pending** (red), Payment status (pending / partially_paid / paid / overdue), GST badge.
- **Filters** — Payment status, GST (yes/no).
- **New Sale fields** — **Unit** (only units currently **available** are listed), Customer name, Customer mobile, Sale amount, Amount received, Booking date, Sale date, **GST applicable (5%)** checkbox, GST %, Other charges, Status (booked / sold).
- **+ Payment** on any row with a pending balance opens **Collect payment**: amount (defaults to pending), date, mode (online / bank transfer / upi / cheque / cash), includes-GST flag, reference/UTR, plus the full **payment history** for that sale.

Side effect on units: creating a sale flips the unit **units.status** from **available** to **booked** or **sold**; deleting it returns the unit to **available**. Pending collections trigger the **sales_payment_due** notification event.

Action	API	Tables	Side effects
Summary / funnel	GET <code>/api/sales/summary</code> · GET <code>/api/sales/unit-availability</code>	sales + sales_payments, units counts	—
List	GET <code>/api/sales?page=&payment_status=&is_gst=</code>	sales + unit/project joins	—
New sale	POST <code>/api/sales</code>	sales (GST amount computed, pending = amount – received), units.status → booked/sold, audit_logs	notifications sales_payment_due when pending remains
Detail	GET <code>/api/sales/:id</code>	sales + sales_payments history	—
Record payment	POST <code>/api/sales/:id/payments</code> {amount, payment_date, payment_mode, is_gst, reference_number}	sales_payments, sales.amount_received/pending_amount/payment_status updated	Overdue → dashboard & notifications
Export	GET <code>/api/sales/export</code>	CSV, audited	—

Construction ERP
Project Tracking System

MAIN

Dashboard
Projects
Daily Progress
Milestones

SITE MANAGEMENT

Drawings
Issues
Inspections
Test Reports
Documents

RESOURCES & COST

Materials
BOQ
Billing & Cost

Sales

0
Units Sold
18 available

—
Sales Value

—
Collected

—
Pending Collections

Sales register
Export CSV

Unit Sales

+ New Sale

Search

Search...

Payment
All

GST
All

UNIT	CUSTOMER	SALE DATE	AMOUNT	RECEIVED	PENDING	PAYMENT	GST	
A101 Wing A	Nilesh & Rohini Khandelwal 9860112233	2026-08-12	₹72,50,000 incl GST ₹3,62,500	₹72,50,000	—	Paid	GST	Edit
A102 Wing A	Farhan Sheikh 9860223344	—	₹73,20,000 incl GST ₹3,66,000	₹21,96,000	₹51,24,000	Partially Paid	GST	+ Payment Edit
A201 Wing A	Prashant Ambekar 9860556677	—	₹74,15,000 incl GST ₹3,70,750	₹5,00,000	₹69,15,000	Partially Paid	GST	+ Payment Edit

3 records · page 1 of 1

CRM & Sales

4.18 HR & Payroll

The shared employee directory (same records as Admin Console) with leave, salary structures and monthly payroll.

Summary tiles — Active employees, On leave (today), Pending leave requests, Payroll due for the current month (pending net).

Four tabs: **Employees** · **Leave** · **Salary structures** · **Payroll**.

Tab 1 — Employees

- Filters** — search (name, code, email), Department, Status (active / on_leave / resigned / terminated). Pagination.
- Columns** — Code, Name, Designation (+ linked role name), Dept, Project, **Login** (green "Linked · username" badge or gray "No login"), Status, Joined, row actions (Details / Edit / Delete).
- Add / Edit fields** — Employee code, Full name, Email, Phone, **Designation / role** (from Masters → Designations; option labels show the linked login role), **Department** (from Masters → Departments), Date of joining, Date of birth, Gender, Employment type (permanent / contract / probation / intern), Status, Project, Wing, Bank account, PAN, Aadhaar, Address, Active, Remarks.
- Login access block** (in the same modal):
 - *New employee:* "Create login credentials" checkbox (requires `users.create` permission) → temporary password (min 8 chars) + **login role** chips. The role linked to the chosen **designation is added automatically**; other selected roles are retained.
 - *Existing employee with no login:* the same block provisions credentials and links the account (one-to-one; a duplicate link is blocked).
- Employee details modal** (row click) — Profile card (all fields + project/wing), Login access card (account, email, status, masked bank account •••• last4), **Leave balance** table for the current year (type, quota, used, pending, remaining), **Salary history** (effective date, basic, HRA, PF), **Recent leave** and **Recent payroll** tables.







Action	API	Tables	Side effects
List	GET /api/hrms/employees?page=&search=&department=&status=	hrms_employees + projects/wings + linked users + designation→role names	—
Detail	GET /api/hrms/employees/:id	employee + hrms_salary_structures, recent leave & payroll	—
Create	POST /api/hrms/employees (+ optional create_login, login_password, login_role_ids)	hrms_employees; when login requested: users (bcrypt), user_roles (selected + designation-linked role), hrms_employees.user_id set, audit_logs	Shared directory row appears in Admin Console instantly
Edit	PUT /api/hrms/employees/:id	hrms_employees, audit_logs	Login linking same as create (only when not yet linked)
Delete	DELETE /api/hrms/employees/:id	hrms_employees — linked login is retained but unlinked	audit_logs
Departments / designations	GET /api/hrms/departments · GET /api/hrms/designations	hrms_departments, hrms_designations (managed from Admin Masters)	—
Login roles list	GET /api/hrms/login-roles	active roles	—

BuildTrack
Project Tracking · v1.0

HR & Payroll

Search projects, tasks, materials...

AT Anupam Tiwari (Super Admin)

Active employees 47

On leave today 3

Pending leave requests 5

Payroll · Aug 2026 ₹18.4 L

Employees | Leave | Salary structures | Payroll

Department: Status:

Code	Name	Designation	Dept	Project	Status	Joined	Actions
BT001	Anupam Tiwari · Super Admin	Indian	Project A	Active	11 Aug 2026	Edit	
BT045	Rajesh Sharma · Project Manager	Engineer	Project A	Active	17 Jun 2026	Edit	
BT112	Sneha Kulkarni · Civil Engineer	Engineer	Project	On Leave	21 Jun 2026	Edit	

+ Add employee

HR & Payroll

Tab 2 — Leave

- **List** — Employee, Type (code badge), From, To, Days, Status (pending / approved / rejected). Filters: status, search.
- **New request** — Employee (active only), Leave type (from Masters → Leave types with quota & paid flag), From, To (days computed), Reason.
- **Approve / Reject** via PUT /api/hrms/leave-requests/:id/decide (requires hrms.approve; Admin Console shows the same workflow). Approved **unpaid** leave feeds payroll loss-of-pay.

Action	API	Tables
List	GET /api/hrms/leave-requests?status=&search=	hrms_leave_requests + employee + type
Create	POST /api/hrms/leave-requests	hrms_leave_requests (total_days computed), audit_logs
Decide	PUT /api/hrms/leave-requests/:id/decide	hrms_leave_requests (status, decided_by, decided_on), audit_logs

Tab 3 — Salary structures

- **Columns** — Employee, Dept, Effective, Basic, HRA, **Gross** (basic+HRA+DA+special+other, computed in UI), PF (ee), Edit/Delete.

Fields — Employee, Effective from, Basic, HRA, DA, Special, Other, PF (employee/employer), ESIC (employee/employer), Professional tax, Remarks.

- An employee's **latest active structure** (effective $\leq$ end of payroll month) is the one payroll uses.

Action	API	Tables
CRUD	GET POST PUT DELETE /api/hrms/salary-structures...	hrms_salary_structures, audit_logs

Tab 4 — Payroll

- **Month picker** (defaults to current month) + **Generate monthly payroll** button (hrms.create).
- **Columns** — Employee, Dept, **Paid days** (1 dp), Gross, Deductions, **Net**, Payment status (pending / partial / paid), **Record payment** action.
- **Record payment modal** — Amount paid ($\leq$ net), payment date, Reference/UTR. Status is derived: 0 $\rightarrow$ pending, $<$ net $\rightarrow$ partial, $\geq$ net $\rightarrow$ paid. A month that already has money paid **cannot be regenerated**.

Payroll calculation (server-side, per employee for the month):

```

structure = latest active salary structure for the employee
total    = working days in the month
lop      = approved days of UNPAID leave types falling in the month
paid     = total - lop                (ratio = paid / total)
gross    = (basic + HRA + DA + special + other)  $\times$  ratio
deduct   = min(gross, PF-ee  $\times$  ratio + ESIC-ee  $\times$  ratio + prof tax)
net      = gross - deduct
 $\rightarrow$  hrms_payroll row (payment_status = pending)

```

Bulk generation processes all active employees; those without a structure (or with blocked re-generation) are reported as **skipped** in the toast.

Action	API	Tables
List month	GET /api/hrms/payroll?payroll_month=&page=	hrms_payroll + employee
Generate (one)	POST /api/hrms/payroll/generate	hrms_leave_requests (unpaid approved), hrms_salary_structures $\rightarrow$ hrms_payroll insert/update
Generate (bulk)	POST /api/hrms/payroll/generate-bulk {payroll_month}	same, all active employees; skips returned
Record payment	PUT /api/hrms/payroll/:id/payment	hrms_payroll (paid_amount, date, reference, derived payment_status)
Summary tiles	GET /api/hrms/summary	hrms_employees, hrms_leave_requests, hrms_payroll

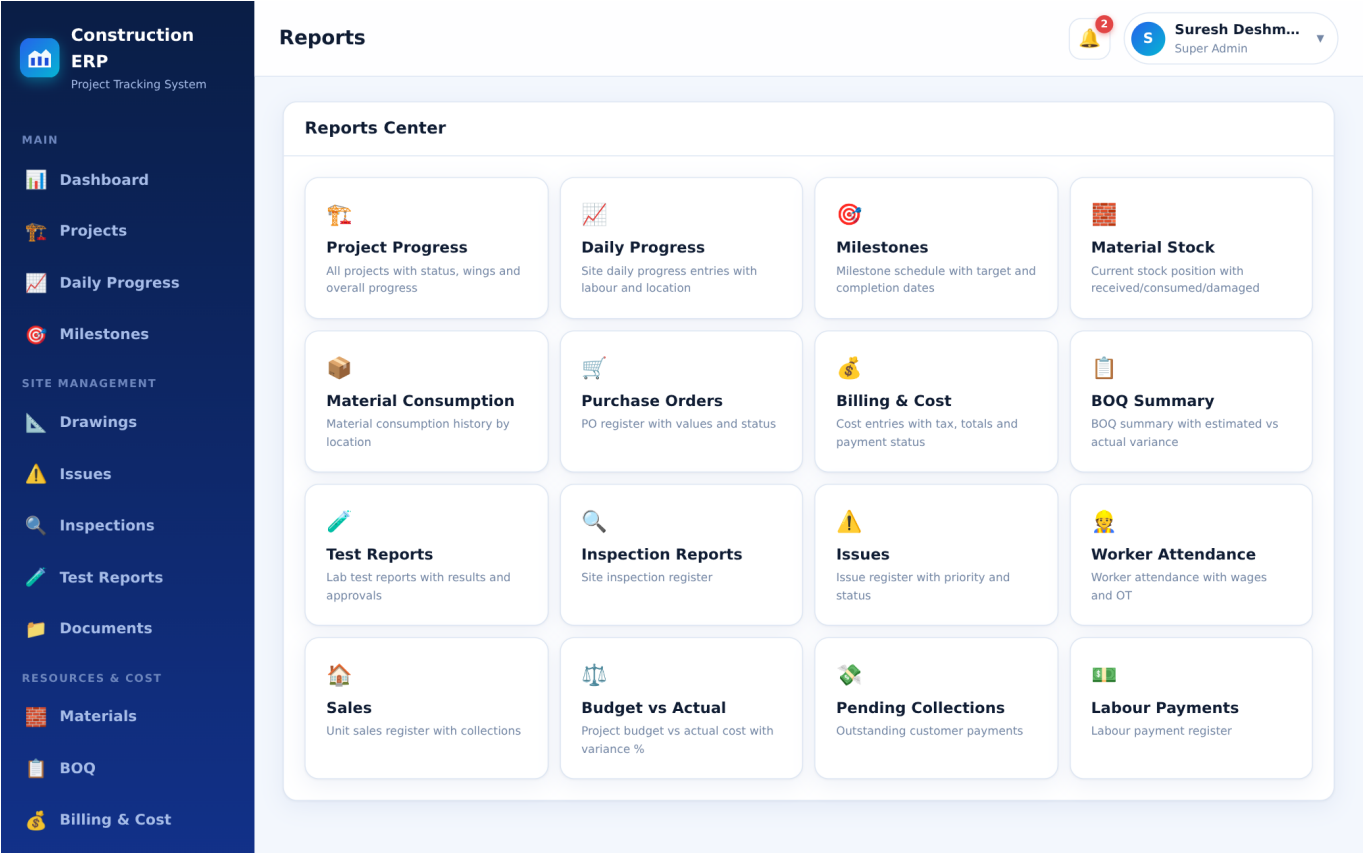
4.19 Reports & Dashboards

A self-service reports center — no SQL needed.

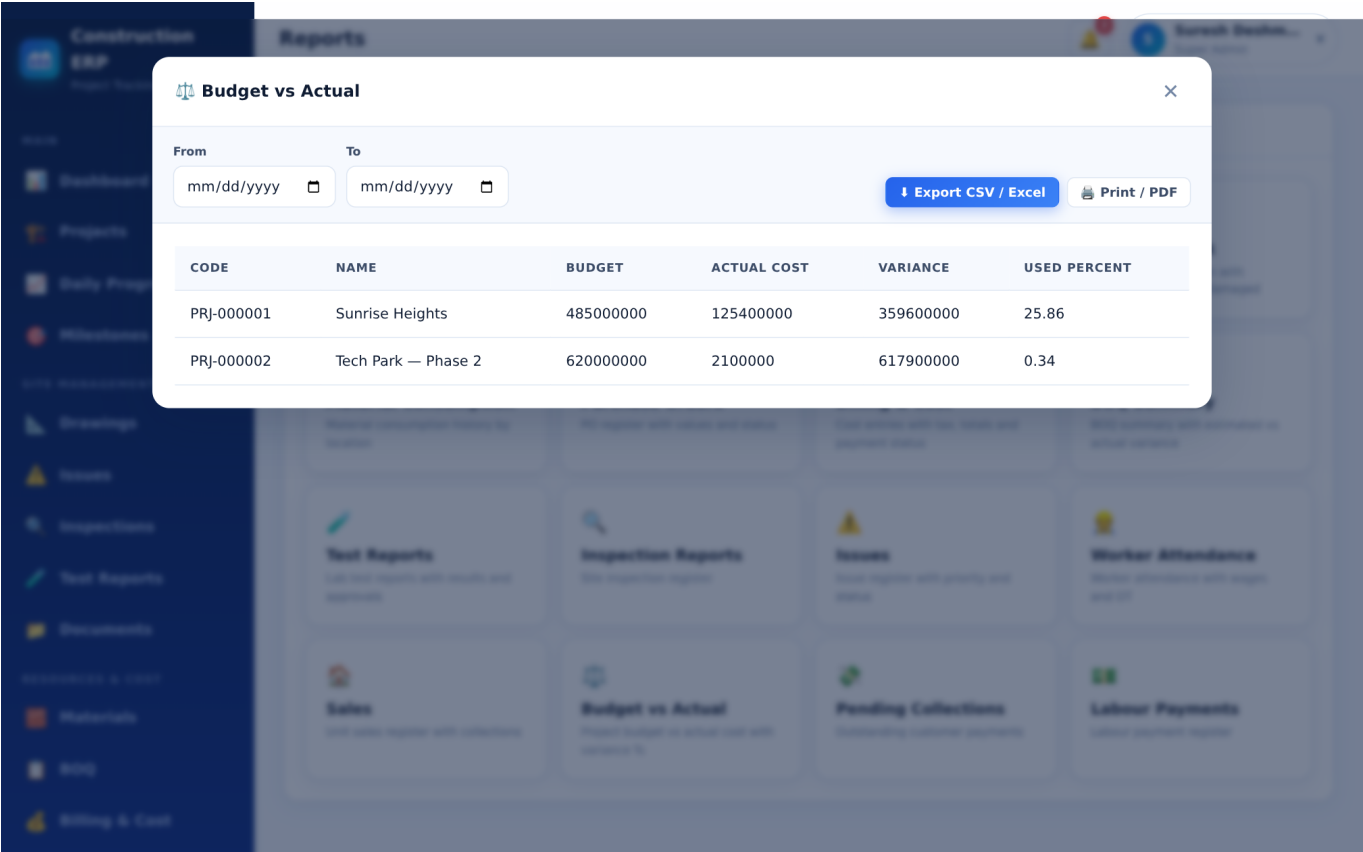
- **17 report cards** (grouped by domain): Project Progress, Daily Progress, Milestones, Material Stock, Material Consumption, Purchase Orders, Billing & Cost, BOQ Summary, Test Reports, Inspection Reports, Issues, Worker Attendance, Sales, Budget vs Actual, Pending Collections, Labour Payments, Project Cost.
- **Click a card** $\rightarrow$ viewer with optional **From/To** date range, the data table (first 12 columns shown), **Export CSV/Excel** and **Print / PDF** (browser print).

UI	API
Report list	GET /api/reports $\rightarrow$ [{key, label}]
View	GET /api/reports/:key?from=&to= $\rightarrow$ {label, data:[...]}
Export	GET /api/reports/:key?format=csv&from=&to= (streamed CSV)

Each report is a read-only query over the tables documented in the screen sections above (e.g. **budget-vs-actual** joins **projects** with **cost_entries**; **material-stock** uses the same ledger aggregation as the Stock tab).



Reports center



Report viewer with export

4.20 My Profile

Covered in §3.3 — reachable from the user chip in the Web Portal. Includes the HR link card and the project-access list for scoped users.

The screenshot displays the 'My Profile' interface. On the left is a dark blue sidebar with the 'Construction ERP' logo and a list of modules: MAIN (Dashboard, Projects, Daily Progress, Milestones) and SITE MANAGEMENT (Drawings, Issues, Inspections, Test Reports, Documents). Below these are 'RESOURCES & COST' (Materials, BOQ, Billing & Cost). The main content area is titled 'My Profile' and shows the user's profile card for Suresh Deshmukh (admin@constructionerp.com · EMP-001) with roles 'Super Admin' and 'SUPER ADMIN'. Fields for 'Full name' (Suresh Deshmukh) and 'Phone' (9822001100) are present, along with a 'Save profile' button. A note states: 'As Super Admin you have access to all projects and modules.' To the right is a 'Change Password' section with input fields for 'Current password', 'New password (min 8 chars)', and 'Confirm new password', and a 'Change password' button. The top right corner shows a notification bell with 2 alerts and a user dropdown for Suresh Deshmukh (Super Admin).

My Profile

4.21 In-app notifications (Web Portal)

Covered in §4.0. Event types raised by the backend: `issue_created`, `issue_assigned`, `issue_updated`, `issue_overdue`, `milestone_delayed`, `po_approval`, `material_received`, `material_shortage`, `test_failed`, `inspection_failed`, `drawing_revision`, `document_expiry`, `sales_payment_due`. Each is individually toggleable in Admin Console → Masters → Notification Events, and each notification carries a `module` + `record_id` used for the deep-link routing.

5. Admin Console — complete screen & tab guide

The Admin Console (separate login on its own port/URL) is where platform administrators manage **who** can do **what** across **which projects**, keep master data consistent, and watch the system.

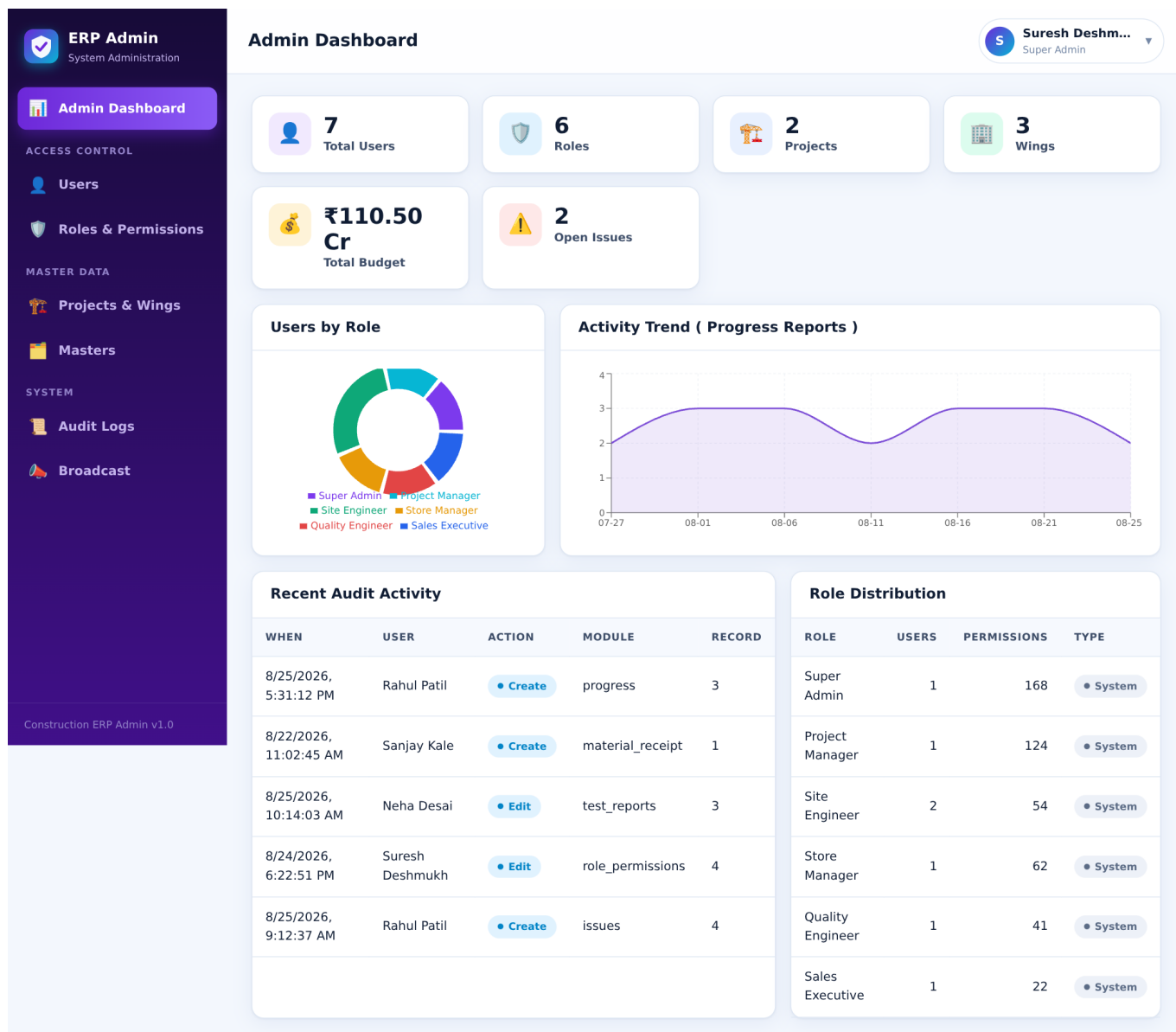
5.0 Layout & navigation

- **Sidebar groups:**
 - (top) **Admin Dashboard**
 - **Access Control** — Users & Employees (`users.view`), Roles & Permissions (`roles.view`)
 - **Master Data** — Projects & Wings (`projects.view`), Masters (visible to all admins)
 - **People & finance operations** — HR & Payroll (`hrms.view`), Petty Cash (`petty_cash.view`)
 - **System** — Audit Logs (`admin.view`), Broadcast (`notifications.create`)
- **Top bar** — page title, "Admin Console · v1.0" chips, user chip with Logout.
- Every route is guarded by `Require(perm)`; the server re-checks the same permissions.

5.1 Admin Dashboard

- **KPI tiles** — Total Users, Roles, Projects, Wings, Total Budget (compact ₹), Open Issues.
- **Users by Role** donut (role → user count), **Activity Trend** (progress reports per day, last 30 days).
- **Recent Audit Activity** table (last 8 events: when, user, action, module, record).
- **Role Distribution** table (role, users, permissions, System/Custom type).

UI	API	Tables
KPIs + charts	GET /api/dashboard/overview	same aggregates as the web dashboard
Users / roles	GET /api/users?limit=100 · GET /api/roles	users (+ user_roles, roles.user_count), roles (+ permission_count)
Recent audits	GET /api/admin/audit-logs?limit=8	audit_logs



Admin Dashboard

5.2 Users & Employees

Manage login accounts **and** their HR/payroll identity in one place.

- **Columns** — User (avatar, name, email), Employee code, Phone, **Roles** (badges), **HR / Payroll** (green "Linked · EMP-xxx" or gray "No employee"), Last login (or "never"), Status.
- **Filters** — search (name, email, code, phone), Role. Pagination. ↓ **Export CSV** (`users.export`).
- **Row actions** — Edit, **Reset PW**, Delete (never on your own account).

Create / Edit user modal

1. **Account** — Full name, Email (login), Phone, Employee code, **Temporary password** (new users only, min 8 chars), Status (active/inactive).
2. **HR & Payroll link block** —
 - *New user:* "Create employee record" checkbox (requires `hrms.create`) → Department, Designation (linked-role auto-added to the role set), Date of joining, Employment type, Payroll project, Employee status.
 - *Existing user without employee:* "Convert to employee" — same fields; creates the shared-directory row and links it.

3. **Roles** — multi-select chips. Non-super-admins cannot see the **admin/super_admin** roles in the picker (they cannot grant those). The selected designation's linked role is **auto-included** (shown in a hint).
4. **Project Access** — rows of (Project, Wing or "All wings"). Empty list + a non-global role = the user sees **no projects**; Super Admin / Admin roles are global.

Reset password modal — new password (min 8 chars); **all of that user's active sessions are revoked immediately** (`refresh_tokens.revoked_at`).

Action	API	Tables	Side effects
List	GET /api/users?page=&search=&roleId=	users + roles + hrms_employees link + last login	—
Create	POST /api/users {name, email, phone, employee_code, password, status, roleIds, projectAccess, createEmployee?, employee?}	users (bcrypt), user_roles, user_projects (each row project + wing_id 0=all), and when requested: hrms_employees + hrms_employees.user_id back-link	Response meta.employeeId when an employee was created; audit_logs
Edit	PUT /api/users/:id + PUT /api/users/:id/roles (when roles changed) + PUT /api/users/:id/project-access	users, user_roles, user_projects, (employee link fields)	audit_logs with old/new values
Reset password	PUT /api/users/:id/reset-password	users.password, all refresh_tokens revoked	User must re-login everywhere
Delete	DELETE /api/users/:id	users — audit history is preserved	audit_logs
Export	GET /api/users/export	CSV, audited	—

ERP Admin
System Administration

Admin Dashboard

ACCESS CONTROL

Users

Roles & Permissions

MASTER DATA

Projects & Wings

Masters

SYSTEM

Audit Logs

Broadcast

Construction ERP Admin v1.0

User Management

S Suresh Deshm...

Super Admin

Users

Export

+ New User

Search

Name, email, code, phone...

Role

All roles

USER	CODE	PHONE	ROLES	LAST LOGIN	STATUS	
S Suresh Deshmukh admin@constructionerp.com	EMP-001	9822001100	—	never	Active	Edit Reset PW
A Amit Verma pm@constructionerp.com	EMP-002	9822011223	—	never	Active	Edit Reset PW Delete
R Rahul Patil engineer@constructionerp.com	EMP-003	9822033445	—	never	Active	Edit Reset PW Delete
V Vikram Joshi electrical@constructionerp.com	EMP-006	9822055667	—	never	Active	Edit Reset PW Delete
S Sanjay Kale store@constructionerp.com	EMP-008	9822077889	—	never	Active	Edit Reset PW Delete
N Neha Desai quality@constructionerp.com	EMP-011	9822099001	—	never	Active	Edit Reset PW Delete
K Kavita Menon sales@constructionerp.com	EMP-009	9822001122	—	never	Active	Edit Reset PW Delete

7 records · page 1 of 1

Users & Employees

5.3 Roles & Permissions

Define the platform's capability model.

- **Columns** — Role (name + code), Description, **HR designation(s)** (names of designations mapped to this role — useful for spotting role/designation drift), Users count, **Permissions** count badge, Type (System/Custom), Status (Active/Inactive).
- **Row actions** — **Permissions** (open the matrix; the super_admin role itself is read-only), **Delete** (only custom roles **with zero users**).

Permission matrix modal

- Grid: **rows = modules**, **columns = the 8 actions** (view · create · edit · delete · approve · export · upload · download). A cell is checkable only when that module.action permission exists in the catalogue.
- Clicking the **module name** toggles the whole module.
- Footer shows "N of M granted" and **Save permissions** writes the full set.

Action	API	Tables	Side effects
Role list	GET /api/roles	roles + user/permission counts + designation names	—
Create role	POST /api/roles	roles, audit_logs	—
Matrix data	GET /api/roles/permissions (catalogue grouped by module) · GET /api/roles/:id (granted ids)	permissions, role_permissions	—
Save matrix	PUT /api/roles/:id/permissions {permissionIds:[...]}	role_permissions (replaced), audit_logs	Takes effect on the users' next request (permissions are loaded per token/me)
Delete	DELETE /api/roles/:id	roles (409 if system or in use)	audit_logs

ERP Admin
System Administration

Admin Dashboard

ACCESS CONTROL

Users

Roles & Permissions

MASTER DATA

Projects & Wings

Masters

SYSTEM

Audit Logs

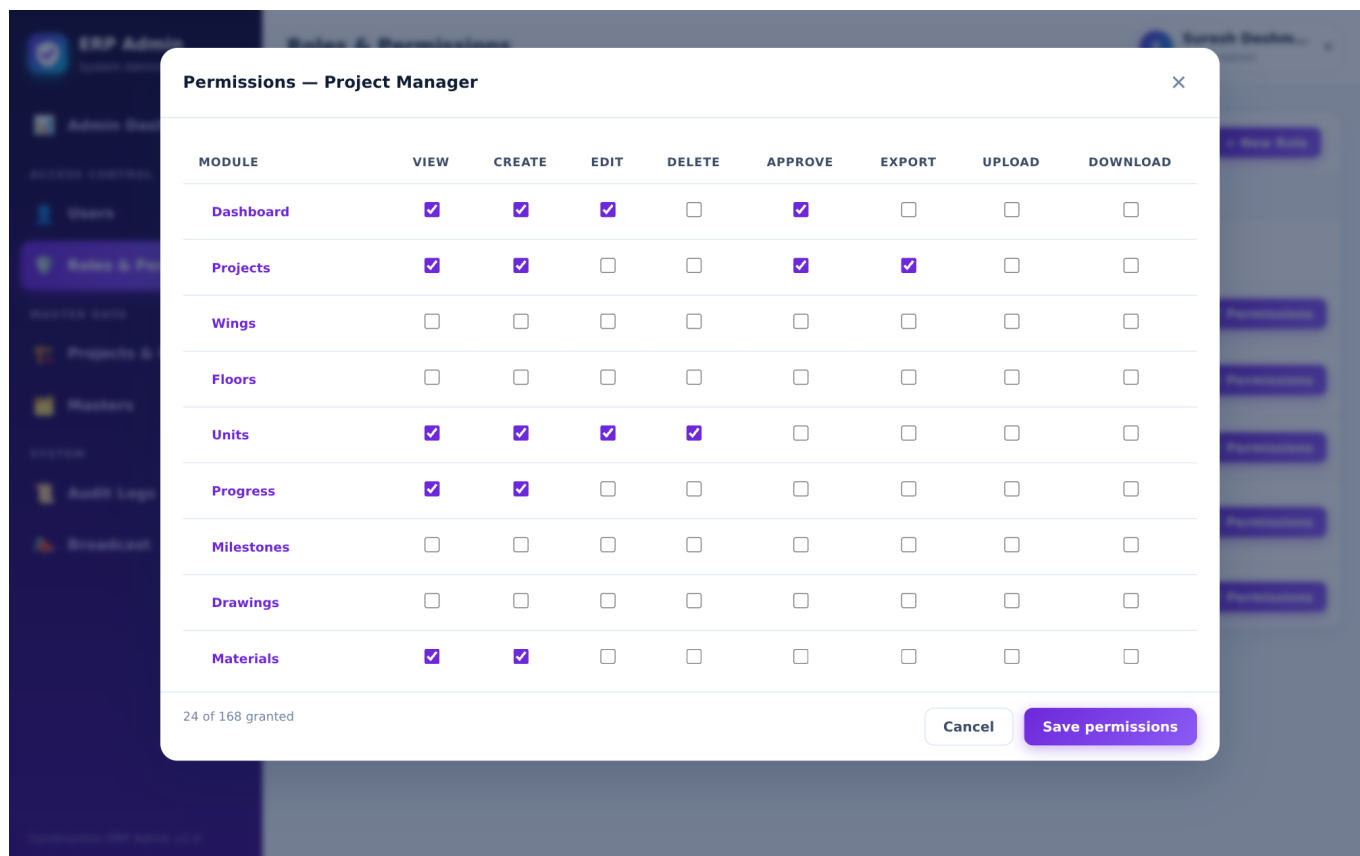
Broadcast

Construction ERP Admin v1.0

Roles & Permissions

ROLE	DESCRIPTION	USERS	PERMISSIONS	TYPE	STATUS	
Super Admin super_admin	Full system access	1	168	System	Inactive	
Project Manager project_manager	Owens assigned projects end-to-end	1	124	System	Inactive	Permissions
Site Engineer site_engineer	Progress, attendance, issues on site	2	54	System	Inactive	Permissions
Store Manager store_manager	Materials, POs, stock	1	62	System	Inactive	Permissions
Quality Engineer quality_engineer	Test reports & inspections	1	41	System	Inactive	Permissions
Sales Executive sales_executive	Unit sales & collections	1	22	System	Inactive	Permissions

Roles & Permissions



Permission matrix

5.4 Projects & Wings (admin)

A project-structure console for admins — the same **projects** / **wings** / **floors** tables the portal uses.

- **Projects table** with inline **New Project** (same fields as the portal's New Project modal) and row-level **Edit** / **Delete** / **Expand**.
- **Expanded row** shows the project's **Wings** table (add/edit/delete wing with progress + status) and its **Floors** table (per wing: floor, status) — so an admin can set up or repair the hierarchy without touching the portal.

Action	API	Tables
Projects CRUD	GET POST PUT DELETE /api/projects...	projects, audit_logs
Wings CRUD	GET POST PUT DELETE /api/wings?projectId=	wings, audit_logs
Floors	GET /api/floors?projectId=&wingId= (+ update)	floors, audit_logs

ERP Admin
System Administration

Admin Dashboard

ACCESS CONTROL

Users

Roles & Permissions

MASTER DATA

Projects & Wings

Masters

SYSTEM

Audit Logs

Broadcast

Construction ERP Admin v1.0

Projects & Wings

Suresh Deshm...
Super Admin

Projects (2)

CODE	PROJECT	CITY	WINGS	STATUS	PROGRESS	BUDGET	MANAGER
PRJ-000001	Sunrise Heights	Nagpur	—	In Progress	38%	₹48,50,00,000	Amit Verma
PRJ-000002	Tech Park — Phase 2	Nagpur	—	Planning	2%	₹62,00,00,000	Amit Verma

Projects & Wings admin view

5.5 Masters (11 tabs)

Master data that feeds dropdowns across every app. Tab visibility is permission-gated (e.g. HR tabs need **hrms.view**).

Tab	Manages	Key fields / behaviour
Departments	hrms_departments	Name, Code (auto from name if blank), Description, Active. Feeds employee forms in both apps.
Designations / Roles	hrms_designations	Name, Code, Linked role (select from login roles — required), Description, Active. The linked role is auto-granted when a login is provisioned from an employee with this designation, and is shown as the "effective designation" on profiles.
Material Categories	material_categories	Name, Description, Consumable flag, Active. Feeds Materials master.
Labour Categories	worker_categories	Name, Base daily rate (₹), Description, Active. Feeds Workers.
Labour Rates	labour_rates	Worker category, Contractor (optional), Daily rate, OT rate/hr, Effective from (date-based rate cards).
Contractors	contractors	Name, Contact, Phone, Email, GST, Address. Feeds Workers.
BOQ Categories	boq_categories	Name, Description, Active. Feeds BOQ items.
Test Types	test_types	Name, Description/standard, Active. Feeds Test Reports.
Inspection Types	inspection_types	Name, Checklist template (items separated by ; — becomes the pre-filled inspection checklist), Active.
Issue Categories	issue_categories	Name, Description, Active. Feeds Issues.
Notification Events	notification_settings	Event key, Label, Enabled toggle — the switch for every event-driven notification (issue_overdue, low stock, document_expiry, ...). Disabled = event silently skipped.

Each tab is a standard CRUD table (search, create, edit, delete, export where configured):

Tab	API base	Table
Departments	/api/hrms/departments	hrms_departments
Designations	/api/hrms/designations	hrms_designations
Material Categories	/api/materials/categories	material_categories
Labour Categories	/api/worker-categories	worker_categories

Tab	API base	Table
Labour Rates	/api/labour-rates	labour_rates
Contractors	/api/contractors	contractors
BOQ Categories	/api/boq/categories	boq_categories
Test Types	/api/test-types	test_types
Inspection Types	/api/inspection-types	inspection_types
Issue Categories	/api/issues/categories	issue_categories
Notification Events	/api/notifications/settings	notification_settings

On an existing installation, run `npm run migrate + npm run seed` in the backend so the new masters are created and legacy employee text values are promoted into managed entries.

The screenshot displays the 'ERP Admin' interface. On the left is a dark sidebar with navigation links: Admin Dashboard, Users, Roles & Permissions, Projects & Wings, Masters (highlighted), Audit Logs, and Broadcast. The main area is titled 'Master Data' and shows a tabbed interface with 'Material Categories' selected. Below the tabs is a search bar and a table with columns: NAME, DESCRIPTION, TYPE, and ACTIVE. The table lists five categories: Steel, Cement, Aggregates, Masonry, and Finishing, all with 'Non-consumable' type and 'Active' status. Each row has 'Edit' and 'Delete' buttons. At the bottom right, it shows '5 records · page 1 of 1' with navigation arrows. The top right corner shows the user 'Suresh Deshm...' as 'Super Admin'.

Masters console

5.6 HR & Payroll (admin operations view)

An operational console over the **same** employee data the Web Portal edits — identity, login coverage, leave workflow and payroll settlement at a glance. Access chip shows **Operational** vs **Read-only** based on `hrms.ed` `it/hrms.create`.

- **KPI tiles** — Active employees, Linked logins (count of employees with a login), Pending leave, **Payroll due** for the month (pending net).
- **Employee identity & login coverage** — searchable, paginated table: Code, Employee, Designation (+ linked role), Department, Project, **Login** (Linked · name / No login), Status. Note: the list is explicitly shared — "Shared with the Project Tracking HR & Payroll panel".
- **Recent leave workflow** — latest 8 requests with **Approve** / **Reject** buttons on pending rows (`hrms.approve`).
- **Payroll payment status** — latest 8 payroll rows for a chosen month (Gross, Net, Paid, status) with **Record payment** (same modal as the portal: amount ≤ net, date, reference) and a **Generate payroll** button (bulk, same engine as the portal).

Action	API	Tables
Summary	GET /api/hrms/summary	hrms_employees, hrms_leave_requests, hrms_payroll
Employees	GET /api/hrms/employees?page=&search=	hrms_employees + links

Action	API	Tables
Leave decide	PUT /api/hrms/leave-requests/:id/decide	hrms_leave_requests
Payroll list	GET /api/hrms/payroll?payroll_month=	hrms_payroll
Generate	POST /api/hrms/payroll/generate-bulk	hrms_payroll (engine of \$4.18)
Record payment	PUT /api/hrms/payroll/:id/payment	hrms_payroll

The full create/edit/delete of employees, salary structures and leave types is done in the **Web Portal → HR & Payroll** (or from Users → "Create employee record" here); this admin view is for the settlement workflow.

5.7 Petty Cash (admin view)

The same ledger as the portal (identical data & API), presented for finance oversight: project filter, type/category filters, summary (cash on hand, spend by category, recent activity), entries table, CSV export. Entry create/edit/delete uses the same **petty_cash.*** permissions.

Action	API	Tables
List / summary / export / CRUD	GET /api/petty-cash · GET /api/petty-cash/summary · GET /api/petty-cash/export · POST PUT DELETE /api/petty-cash...	petty_cash_entries, audit_logs

5.8 Audit Logs

The complete mutation trail of the platform — every create, update, delete and export by every user.

- **Columns** — When (timestamp), User, Action badge (create green / update blue / delete red / export purple), Module, Record id, IP.
- **Filters** — search (user or record), Module (all 26 audited modules: auth, users, roles, projects, wings, floors, units, progress, milestones, drawings, materials, inventory, purchase_orders, billing, boq, test_reports, inspections, issues, workers, attendance, labour_payments, sales, documents, reports, notifications, system), User (directory select).
- **Row click → detail modal** — full JSON of **old value** and **new value**, plus IP and user-agent.
- **↓ Export CSV** (admin.export).

Action	API	Tables
List	GET /api/admin/audit-logs?page=&module=&userId=&search=	audit_logs (indexed by module/user/time)
Export	GET /api/admin/audit-logs/export	CSV, audited

How entries are created: every controller mutation calls the shared **audit()** helper *inside the request*, writing user_id, user_name, action, module, record_id, old/new JSON, IP and user-agent. Auditing is fire-and-forget — it can never fail the business operation.

5.9 Broadcast

Push an announcement to specific users or the whole organisation.

- **Fields** — Title, Message, **Type** (info / success / warning / error), **Audience:** "All users" (default) or an individual user picker.
- **Send** writes one **notifications** row per target user (title, message, type, module=system). Target users see it in their bell next poll; mobile users on registered devices also receive it via the Expo push relay using their stored push token.
- **Sent count** is shown after each broadcast.

Action	API	Tables	Side effects
User picker	GET /api/auth/user-directory	users (active)	—
Send	POST /api/notifications/broadcast {title, message, type, userIds?[]}	notifications (one row per user)	Push tokens (user_push_tokens) relayed where available

6. Mobile App — screen guide (site engineer)

The Expo (React Native) app is the field companion: camera + GPS capture that works **offline** and syncs automatically.

6.1 Screens

Screen	Purpose
Login	Email + password against the same API (POST /api/auth/login); tokens stored in SecureStorage; forgot-password link
Dashboard	Quick stats for the engineer's projects (open issues, pending inspections, recent progress) + quick actions
Progress List	Daily reports for the engineer's projects, with sync-state indicators
Progress Capture	The star screen: project/wing/floor, date, work description, %, labour, weather, remarks, camera photos (each auto-tagged with GPS + timestamp), GPS capture, save — works offline
Progress Detail	Full report with photos (each showing source + coordinates)
Attendance	Mark worker attendance for a project/date (same statuses as web: present/absent/half day/leave/overtime + OT hours)
Inspections / Inspection Detail	Checklist inspections with pass/fail/N-A, GPS, photos
Issues / Issue Create / Issue Detail	Raise and resolve issues with photos + GPS; comment thread
Notifications	In-app notification list (polling + Expo push), mark read
Sync Queue	Every offline item with its state (queued / syncing / error / done), retry buttons
More	Profile, device push-token registration, logout, app info

6.2 Offline queue (how it works)

```
capture (no network)
  | client_ref = unique id generated on device
  ↓
AsyncStorage '@cerp/offline-queue' (idempotent: same client_ref never enqueued twice)
  | background auto-sync + manual "Sync now"
  ↓
POST /api/progress/sync (batch) or POST /api/issues / POST /api/attendance
  |
  | 200 created → item removed from queue
  | duplicate → server returned the existing row id (client_ref guard) → item removed
  | error → kept in queue, retried with backoff, visible in Sync Queue screen
```

- The web app uses the same `client_ref` guard for its own offline fallback queue.
- Push notifications: the app registers its Expo push token once via `POST /api/notifications/push-token` (stored in `user_push_tokens`) and relays in-app notifications through the standard endpoint.

Action	API	Tables
Create progress (online)	POST /api/progress (multipart, per-photo GPS + meta)	daily_progress, progress_photos, file_uploads + roll-up
Batch sync (offline items)	POST /api/progress/sync	as above; per-item created/duplicate/error results
Attendance	POST /api/attendance	attendance (server-computed wages)
Inspections	POST /api/inspections	inspections, inspection_items
Issues	POST /api/issues (multipart)	issues, photos, notifications
Push token	POST /api/notifications/push-token	user_push_tokens
Notifications	GET /api/notifications · PUT /api/notifications/:id/read	notifications

6.2b Marketing site

The public landing page (`marketing/`) is a static React site with no authentication. It describes the ERP's actual capabilities — project structure, daily progress with geo-tagged photos, material lifecycle, BOQ variance, quality, workforce, HR & payroll, petty cash, equipment, sales, notifications, audit — with module names that match the real applications. It is for prospects and is not part of the operational data flow.

7. End-to-end data flows (complete)

This chapter chains the per-screen flows into the full cross-module journeys. Each flow lists the tables touched in order and the side effects.

7.1 Authentication & session lifecycle

```
login(email,pw) → POST /auth/login
    |
    |— users (bcrypt check, last_login_at)
    |— refresh_tokens (new hashed token)
    |— JWT access (id, roles, permissions) + refresh
every request → Authorization: Bearer <access>
    |
    |— middleware authenticates (JWT verify)
    |— attachProjectScope (user_projects → project/wing sets)
    |— requirePermission(module.action)
token expiry → POST /auth/refresh (old refresh revoked, new issued)
logout/reset PW → POST /auth/logout | PUT /users/:id/reset-password
    |
    |— refresh_tokens.revoked_at = now (all devices out)
```

Actor	Path
Any user, any app	Login → tokens → scoped, permission-checked requests → transparent refresh → revocation on logout/reset

7.2 User & employee provisioning (identity sync)

Two entry points, one identity:

```
A) Admin Console → Users → New User
  POST /users → users (bcrypt) → user_roles → user_projects
  [+ createEmployee?] → hrms_employees (+ user_id back-link)
B) Web Portal → HR & Payroll → Add employee
  POST /hrms/employees → hrms_employees
  [+ create_login?] → users → user_roles (selected u designation-linked role)
    |
    |— hrms_employees.user_id = new user id
```

- **One-to-one link:** a user and an employee can never be double-linked (both UIs block it; the server rejects it).
- **Designation → role mapping:** `hrms_designations.role_id` means "employees with this designation automatically get this login role" — enforced at provisioning, displayed as the **effective designation** on profiles.
- **Delete semantics:** deleting the **employee** keeps the login (unlinked); deleting the **user** keeps the audit history.
- **Audit:** every step writes `audit_logs` rows.

7.3 Project setup & progress roll-up

```
Admin/PM creates project → projects
  adds wing → wings (manager, dates, status)
  adds floors → floors (per wing)
  adds units → units (saleable area, price, status=available)

daily progress saved (anywhere: web, mobile, sync)
  ↳ recomputeProjectProgress(projectId) [same DB transaction]
    wing.progress = AVG(daily_progress.percentage) per wing
    projects.overall = AVG(wing.progress) per project
```

Tables: projects, wings, floors, units, daily_progress.

Consumers of roll-up: Dashboard (KPI + 30-day trend + status donut), Project Detail header, Reports (project-progress), notifications on delayed milestones.

7.4 Daily progress entry — web & mobile (with photos, GPS, offline)

#	Step	UI	API	Tables written	Side effects
1	Fill report (project→wing→floor, date, %, labour, weather, text)	Portal *New Report* / Mobile *Capture*	—	—	GPS captured (browser / device)

#	Step	UI	API	Tables written	Side effects
2	Attach photos (≤8)	file pick / camera	—	—	per-photo GPS + timestamp in photosMeta
3	Save online	both	POST /api/progress (multipart, client_ref)	file_uploads, progress_photos, daily_progress	progress roll-up; audit
4	Save offline (mobile)	queue	(none)	device storage @cerp/offline-queue	item visible in Sync Queue
4b	Auto-sync	background	POST /api/progress/sync	as step 3, per item	duplicate guard by client_ref; per-item result
5	View / deep-link	bell or list	GET /api/files/:id per photo	file_uploads read	photo shows source + coords
6	Dashboard reflects it	—	GET /api/dashboard/overview	—	trend + recent-updates lists

7.5 Material procurement cycle (requirement → PO → GRN → stock)

#	Step	Actor	API	Tables	Side effects
1	Engineer raises requirement	Site	POST /materials/requirements	material_requirements (REQ-...)	—
2	Store approves / rejects	Store Mgr	PUT /materials/requirements/:id/status	material_requirements	notification po_approval to requester
3	Store creates PO (items, rates, tax, discount)	Store Mgr	POST /materials/purchase-orders	purchase_orders (auto totals) + purchase_order_items	status draft
4	Submit → Approve → Mark sent	Store Mgr / PM	PUT /materials/purchase-orders/:id/status	purchase_orders (+approved_by/at)	approval notifies creator
5	Delivery received — GRN	Store	POST /materials/receipts	material_receipts + material_receipt_items (accepted = received – damaged)	stock_transactions (+ accepted); PO item received_qty; PO → partially_received/received; notification material_received
6	Stock position updates live	Store/PM	GET /materials/stock	ledger aggregate	LOW STOCK badge when ≤ min level (dashboard KPI + material_shortage event)
7	Optional returns	Store	POST /materials/returns → PUT .../approve	material_returns + ledger (+qty)	stock back up
8	Accounts settles vendor	Account	POST /billing/cost-entries (type consumable, supplier, bill no)	cost_entries	vendor-due KPI, budget-vs-actual

7.6 Material consumption on site

```

engineer → POST /materials/consumption {project, wing, material, qty, date, location}
├─ read current stock (Σ stock_transactions for project+material)
├─ reject if qty > stock
├─ insert material_consumption + stock_transactions (-qty)
└─ if stock now ≤ min level → low-stock notification

```

Tables: material_consumption, stock_transactions. **Feeds:** Stock tab ledger, Material Consumption report, project cost (when mirrored as billing entries), dashboard.

7.7 Workforce wage cycle (worker → attendance → payment)

#	Step	UI	API	Tables	Computation
1	Register worker (category, contractor, wage, OT rate)	Workers tab	POST /workers	workers	—
2	Mark attendance per day (status + OT hours)	Attendance tab / Mobile	POST /attendance (batch upsert)	attendance	daily_amount = wage × status factor (1, ½, 0); overtime_amount = OT hrs × OT rate — server-side
3	Summary & payroll preview	badges	GET /attendance/summary	aggregate	day total

#	Step	UI	API	Tables	Computation
4	Generate payment for a period	Labour Payments	POST /labour-payments	labour_pay	gross = $\Sigma(\text{daily} + \text{OT})$ over period; net = gross – deductions
5	Settle	Mark paid	PUT /labour-payments/:id	labour_pay	status paid/partial; payment date + mode
6	Monthly wage report	Reports / API	GET /attendance/monthly-report	aggregate	days, OT, gross, paid vs pending per worker

Feeds: Dashboard "Pending Labour Payment" KPI, **labour-payments** report.

7.8 HR payroll cycle (structure → leave → bulk payroll → payment)

```

1. HR sets salary structure → hrms_salary_structures (basic/HRA/DA/..., PF, ESIC, PT, effective_from)
2. Employee raises leave → hrms_leave_requests (total_days computed)
3. Manager decides → PUT .../decide (approved/rejected, decided_by)
4. End of month: Generate → POST /hrms/payroll/generate-bulk {payroll_month}
   per active employee:
       structure = latest active with effective_from ≤ end of month
       lop       = approved days of UNPAID leave types in month
       paid      = working_days - lop ; ratio = paid / working_days
       gross     = (basic+HRA+DA+special+other) × ratio
       deduct    = min(gross, (PF-ee + ESIC-ee) × ratio + PT)
       net       = gross - deduct
       → hrms_payroll (payment_status = pending) [skips reported in response]
5. Accounts records payment → PUT /hrms/payroll/:id/payment
   status: 0=pending | <net=partial | ≥net=paid (paid months are locked from re-generation)

```

Tables: **hrms_salary_structures**, **hrms_leave_requests**, **hrms_leave_types**, **hrms_payroll**. **Feeds:** HRMS summary tiles, payroll KPIs, admin payroll status view, employee detail modal.

7.9 Unit sales & collections cycle

```

units.status = available (created under project structure)
|
| sales exec: New Sale (unit only listed if available)
|
▼
POST /sales → sales (amount, GST, received, pending) + units.status → booked|sold
|
| customer pays (staged)
|
▼
POST /sales/:id/payments → sales_payments + sales.amount_received/pending/payment_status
|                                     pending>0 → notification sales_payment_due
|
▼
KPIs: units sold · value · collected · pending      Reports: sales · pending-collections
deleting a sale returns the unit to available

```

7.10 Equipment operation & hire billing cycle

```

1. register equipment → equipment (type, ownership, rates, project, status)
2. daily log → POST /equipment/logs
   equipment_logs (+hours, fuel, status_after)
   equipment.status = status_after (fleet board always current)
3. month end → POST /equipment/billing/generate {billing_month}
   hours =  $\Sigma$  running_hours of the month
   amount = hours × hourly_rate → equipment_billing (pending)
4. settlement → PUT /equipment/billing/:id (paid_amount, status)

```

Feeds: equipment summary tiles (fleet by status, month hours + fuel), billing entries in finance, project cost reports.

7.11 Quality assurance flow

```

Material test:  lab result → POST /test-reports (file, values, result)
                result=fail → notification test_failed
                submitted → PUT /test-reports/:id/status (approve|reject)

Inspection:    POST /inspections {type, items[], GPS}
                items → inspection_items (pass/fail/N-A)
                any fail → status failed + notification inspection_failed
                passed/failed/reinspection_required → PUT ... (close)

```

Tables: **test_types**, **test_reports**, **file_uploads**, **inspection_types**, **inspections**, **inspection_items**.

7.12 Issue lifecycle & notifications

```

raise (ISS-####, photos, GPS, assignee, due)
|   issue_created → watchers ; issue_assigned → assignee
▼
open → assigned → in_progress → resolved → closed  (reopen allowed)
|   issue_updated → affected users
|   due date passed & unresolved → issue_overdue (red in list)
▼
comment thread (issue_comments) · dashboard open-issues KPI · issues report

```

7.13 Drawing revision control

```

upload drawing (R0) → drawings + drawing_revisions + file_uploads
|   notification drawing_revision → approvers
▼
add revision (R1, R2...) → pending
|
|— approve → drawing.approval_status = approved ; latest revision = current file
|— reject  → drawing.approval_status = rejected
download/preview always via authenticated GET /files/:id

```

7.14 Audit trail & broadcasts (system-wide)

- **Audit:** every mutation in every module → **audit_logs** (user, action, module, record, old/new JSON, IP, agent) → Admin → Audit Logs (filter, inspect, export). Deletes of users/records keep their history.
- **Broadcast:** Admin → Broadcast → **POST /api/notifications/broadcast** → one **notifications** row per target (+ Expo push relay via **user_push_tokens**) → bells in both web apps and mobile list.
- **Notification events** are all gated by **notification_settings.enabled** (Masters → Notification Events).

8. Data dictionary — all 65 database tables

Single MySQL schema (`backend/database/schema.sql`), 65 tables, FKs and indexes included. Grouped by domain.

8.1 Identity & access (9)

Table	Purpose
users	Login accounts: email (unique), bcrypt password hash, name, phone, employee_code, status, last_login_at
roles	Role catalogue: name, code, is_system, is_active
permissions	Permission catalogue: code module.action, module, action
role_permissions	Role ↔ permission join
user_roles	User ↔ role join
user_projects	Project/wing scoping: user, project_id, wing_id (0 = all wings)
refresh_tokens	Rotating refresh tokens: user, sha256 token hash, device info, expires_at, revoked_at
user_push_tokens	Expo push tokens per user (mobile)
password_reset_tokens	Single-use, time-limited reset tokens

8.2 Project structure & progress (10)

Table	Purpose
projects	Project master: code, client, manager, budget, dates, status, overall_progress (computed)
wings	Wings/blocks: code, manager, dates, status, progress (computed)
floors	Floors per wing: sequence, status, progress
units	Saleable units: number, type, saleable area, price, status (available/booked/sold)
milestones	Schedule events: start/target/completion, percentage, responsible, status
daily_progress	Daily site reports: description, %, labour, weather, GPS, client_ref (dedupe)
progress_photos	Photos per report: file ref, per-photo GPS, captured_at, source, uploaded_by
file_uploads	Uploaded files on disk: original name, path, mime, size, owner
drawings	Drawing master: number, category, approval_status, latest revision
drawing_revisions	Revision history: rev no, date, status (pending/approved/rejected), file ref

8.3 Materials & inventory (11)

Table	Purpose
material_categories	Categories with consumable flag
materials	Material master: code, unit, min_stock_level (alert threshold)
suppliers	Vendor master: contact, GST, address
material_requirements	Requirements: qty, needed-by, status (pending→approved/rejected→po_created→fulfilled)
purchase_orders	PO header: supplier, dates, auto totals (subtotal/tax/discount/grand), status, approved_by
purchase_order_items	PO lines: material, qty, unit, rate, tax, amount, running received_qty
material_receipts	GRN header: PO ref, challan, date
material_receipt_items	GRN lines: received, damaged, accepted qty
material_consumption	Consumption records: qty, location, date
material_returns	Returns: qty, status (pending→approved)
stock_transactions	The ledger — every +/- with reference module/id; all stock positions are computed from here

8.4 Finance (6)

Table	Purpose
cost_entries	Vendor cost/billing entries: type, category, tax, paid, payment_status
sales	Unit sales: customer, amount, GST, received, pending, payment_status
sales_payments	Staged collections: amount, date, mode, reference

Table	Purpose
petty_cash_entries	Site cash: topup/expense/replenish, category, paid-to
labour_payments	Worker wage settlements: period, days, OT, gross, deductions, net, paid
equipment_billing	Monthly hire bills: hours, rate, total, payment status

8.5 Workforce (5)

Table	Purpose
contractors	Contractor companies
worker_categories	Labour categories with base daily rate
labour_rates	Date-effective rate cards per category/contractor
workers	Worker master: code, category, contractor, daily_wage, overtime_rate, project/wing
attendance	Daily marks: status, OT hours, computed daily_amount / overtime_amount, shift, marked_by
labour_rates/labour_payments	(see Finance for payments)

8.6 BOQ (3)

Table	Purpose
boq_categories	BOQ item categories
boq	BOQ header: tax %, discount, status, totals
boq_items	Items: est/actual qty × rate, variance

8.7 Quality & issues (8)

Table	Purpose
test_types	Lab test catalogue
test_reports	Results: lab, standard, result values, result_status, approval status, file
inspection_types	Inspection catalogue with checklist template
inspections	Inspection reports: date, inspector, GPS, status
inspection_items	Checklist lines: item, result (pass/fail/N-A)
issue_categories	Issue categories
issues	Issues: priority, assignee, due, GPS, status workflow, photos
issue_comments	Discussion thread

8.8 Documents, equipment & system (6)

Table	Purpose
project_documents	Document vault: category, version, expiry_date
notifications	In-app notifications: user, title, message, type, module, record_id, is_read
notification_settings	Per-event on/off switches
audit_logs	Full mutation trail: user, action, module, record, old/new JSON, IP, agent
equipment	Fleet master: type, ownership, rates, status, project
equipment_logs	Daily hour/fuel logs with status_after

8.9 HRMS (7)

Table	Purpose
hrms_departments	Department master
hrms_designations	Designation master with role mapping (role_id)
hrms_employees	Shared employee directory: all profile fields, project/wing, user_id (login link)
hrms_leave_types	Leave catalogue: code, annual quota, paid flag
hrms_leave_requests	Requests: dates, computed days, status, decision fields
hrms_salary_structures	Salary components per employee with effective_from
hrms_payroll	Monthly payroll: working/paid/Lop days, gross, deductions, net, payment fields

Total: 65 tables. Stock positions, progress percentages, PO totals, payroll figures and payment statuses are

9. Status & workflow reference

Entity	States	Notes
Project	planning · in_progress · on_hold · completed · cancelled	overall_progress auto-computed
Wing	same as project	progress auto-computed
Floor	pending · in_progress · completed · on_hold	pending displays as "planning"
Unit	available · booked · sold	flipped by sales; delete blocked when sold
Milestone	pending · in_progress · completed · delayed	100 % auto-completes
Requirement	pending · approved · rejected · po_created · fulfilled	approve/reject gated by materials.approve
Purchase order	draft · pending_approval · approved · sent · partially_received · received · cancelled	GRN moves it forward automatically
GRN item	received / damaged / accepted	accepted = received – damaged
Material return	pending · approved	approval adds stock back
Cost entry (payment)	unpaid · partial · paid	derived from paid vs total
Labour payment	pending · partial · paid	generated from attendance period
Sale (payment)	pending · partially_paid · paid · overdue	staged payments
Issue	open · assigned · in_progress · resolved · closed	reopen → in_progress; overdue flagged
Inspection	pending · passed · failed · reinspection_required · closed	any failed item → failed
Test report	status: draft → submitted → approved/rejected; result: pending · pass · fail · inconclusive	fail raises notification
Drawing revision	pending · approved · rejected	approval sets drawing approval_status
Equipment	available · deployed · maintenance · breakdown · idle	flips to log's status_after
Equipment billing	pending · partial · paid	monthly, hours × hourly rate
Employee	active · on_leave · resigned · terminated	login link optional
Leave request	pending · approved · rejected	unpaid approved leave feeds payroll
Payroll record	pending · partial · paid	paid months locked from re-generation
User	active · inactive	reset password revokes sessions

10. Tips, FAQ & troubleshooting

General

- *I don't see a menu item.* Your roles don't include the module's **view** permission, or (for scoped modules) you have no matching project assignment. Ask your admin to check Roles → Permissions and Users → Project Access.
- *I can open a screen but a button is missing.* That button needs an extra action (e.g. **approve**, **export**) — same fix.
- *Every screen still blocks me on the server (403).* The API enforces permissions independently of the UI; the message names the exact permission code.

Progress & photos

- *Project % seems "wrong".* It is not typed: wing % = average of that wing's daily progress entries; project % = average of wing %. Delete/correct the underlying reports to change it.
- *Duplicate progress report after mobile sync?* Impossible by design — **client_ref** dedupe; the sync returns **duplicate** with the existing id.
- *Photos show "no GPS".* Geolocation was denied or unavailable at capture; the photo still saves.

Materials

- *Consumption rejected with "insufficient stock".* The ledger has less than the quantity; receive a GRN first or reduce the quantity.
- *Where is my damaged quantity?* In the GRN detail (per item) and the Stock tab's Damaged column — damaged stock never enters available stock.
- *LOW STOCK badge.* Stock fell to or below the material's Min stock level (Masters/Materials field). Raise a requirement.

Workforce & HR

- *Attendance day-pay looks wrong.* Check the worker's daily wage and OT rate on the Workers tab; half day pays 50 %, absent/leave pay 0.
- *Payroll month won't regenerate.* Any month with a recorded payment is locked — correct the payment record instead.
- *Employee missing from payroll generation.* Must be active (status active) and have a salary structure effective in that month; skips are listed in the toast.

Sales

- *Unit not in the New Sale list.* It is already booked/sold (only **available** units are offered).
- *Collections show pending.* Use the row's + **Payment** to record staged payments; status updates automatically.

Security & administration

- *How do I force a user out everywhere?* Users → Reset PW (or the user changes their password) — all refresh tokens are revoked.
- *Who did what to record X?* Audit Logs → filter by module/record; the detail shows full old/new JSON, IP and user-agent.
- *Can I silence a notification type?* Masters → Notification Events → toggle off.

Deployment reminders

- Change all seed passwords; set strong JWT secrets in **backend/.env**; run **npm run smoke** after **db:setup** to verify the 17-point API check; see **docs/INSTALLATION.md** and **docs/DEPLOYMENT.md** for production hardening.